



Bahir Dar University

Bahir Dar Institute of Technology Faculty of Computing

RAG Based Amharic Legal Chatbot

Submitted to the faculty of computing in partial fulfillment of the requirements for the degree
of Bachelor of Science in Software Engineering

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2018

Bahir Dar, Ethiopia

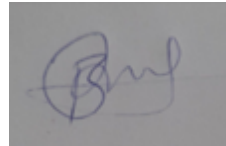
Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

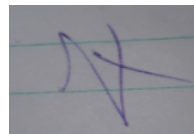
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Roles and responsibilities

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3	Frontend Development (User Interface)		√
4	RAG Pipeline Implementation	√	
5	Vector Database Integration	√	
6	AI Model Integration (LLM)	√	
7	User Authentication & Authorization		√
8	Admin Module Development		√
9	Legal Document Upload & Processing	√	√
10	Database Design & Implementation	√	√
11	Error Handling & Validation	√	√
12	System Testing & Debugging	√	√
13	Performance Optimization	√	
14	Deployment & Configuration	√	√
15	System Maintenance & Updates	√	√

Acknowledgement

We would like to express our sincere gratitude to Bahir Dar University for providing us with the opportunity and resources to carry out this project.

Our deepest appreciation goes to our advisor, Mr. Bzuayehu Tadege, for his invaluable guidance, continuous support, constructive feedback, and encouragement throughout the development of this project. His expertise and patience greatly contributed to the successful completion of this work.

We are also thankful to all instructors and staff members of the university who shared their knowledge and provided assistance during our academic journey.

Finally, we would like to thank our families and friends for their constant moral support and encouragement, which motivated us to complete this project successfully.

List Of Acronyms

Acronym	Full Form
AI	Artificial Intelligence
BDU	Bahir Dar University
NL	Natural Language
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
API	Application Programming Interface
UI/UX	User Interface / User Experience
CSV	Comma-Separated Values
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
EDA	Exploratory Data Analysis
ML	Machine Learning
FREQ	Functional requirement

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Abstract

Access to legal information in Ethiopia is frequently hindered by complex technical terminology and a scarcity of centralized, easily searchable digital resources in local languages. This project presents the development of a **Retrieval-Augmented Generation (RAG)** based Legal Question-Answering System designed specifically for **Amharic**.

The system utilizes a specialized pipeline to preprocess legal corpora, store them in a **NoSQL vector database**, and perform **semantic retrieval** of the most relevant legal provisions. By augmenting the input of a **Large Language Model (LLM)** with these retrieved facts, the system generates accurate, context-aware answers grounded in official proclamations and directives. Key features include **Role-Based Access Control (RBAC)** for Users and Administrators, alongside an automated document management interface for legal experts. By addressing the linguistic nuances of Amharic and the specific requirements of the Ethiopian legal framework, this system aims to democratize legal knowledge, providing citizens and legal professionals with a reliable, efficient, and user-friendly tool for legal inquiry and research.

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Chapter One: Introduction

1.1 Background

The legal system is the foundation of a fair society, yet its effectiveness depends on how easily citizens can understand and access it. In Ethiopia, the legal framework consists of a complex hierarchy of proclamations, regulations, and directives. However, most of these resources remain trapped in scattered PDFs or physical archives, often written in highly technical language that creates a "knowledge gap" between the law and the public.

As Ethiopia undergoes rapid digital transformation, there is a growing need to move beyond traditional, manual consultation processes, which involve long queues and inconsistent advice, toward AI-driven, accessible digital solutions.

1.2 Statement of the Problem

- Despite the foundational importance of the law in Ethiopia, the current system of legal information delivery faces critical failure points that prevent citizens from effectively exercising their rights. These challenges can be categorized as follows:
- **Scattered PDFs and Document Silos**
 - Legal resources are not centralized in machine-readable formats; most proclamations and regulations exist as unindexed images or static PDF files scattered across various ministry websites or private archives.
 - The lack of a unified digital repository makes it nearly impossible for users to perform comprehensive cross-document searches, leading to incomplete or outdated information retrieval.
- **Physical and Geographical Bottlenecks**
 - Many essential legal documents, particularly administrative directives and regional guidelines, are only accessible through physical government offices.
 - For individuals in remote or rural areas, obtaining basic procedural information requires significant travel and time, creating a "justice gap" based on geographical location.
- **Technical Jargon and "Legalese"**
 - The Amharic used in official legal texts is often highly specialized and academic, making it difficult for the average citizen to interpret without professional help.
 - The complexity of the language creates a functional barrier where users may find the correct document but remain unable to understand how the law applies to their specific situation.
- **Inconsistency and AI Hallucinations**
 - Relying on informal advice or generic AI models often leads to misinformation, as standard AI tools are not grounded in the specific, updated statutes of the Ethiopian legal framework.

- Without a system that links answers directly to authenticated sources, users risk making critical decisions based on "hallucinated" data or foreign laws that do not apply locally.
- **Administrative Overload**
 - Government legal officers are frequently overwhelmed by repetitive, basic inquiries from the public, which could be handled through automated systems.
 - The manual nature of current consultation processes leads to long queues, delayed decision-making, and an inefficient use of human legal expertise.

1.3 Proposed System

To address the documented challenges of legal information accessibility in Ethiopia, this project proposes the development of the **Amharic Legal Chatbot**, a specialized AI platform built on a **Retrieval-Augmented Generation (RAG)** architecture. Unlike traditional search engines or standalone AI models, this solution bridges the gap between raw legal data and user-centric legal assistance through a multi-layered technical approach.

- **Retrieval-Augmented Generation (RAG) Architecture**
 - The core of the solution lies in "grounding" the AI. Instead of relying on the pre-existing, static knowledge of a Large Language Model (LLM), which may be prone to hallucinations or outdated information, the RAG pipeline first retrieves relevant text "chunks" from a verified database of Ethiopian laws.
 - This retrieved context is then fed into the LLM as a "fact sheet," ensuring the final Amharic response is based strictly on specific proclamations, regulations, or directives.
- **Specialized Amharic Knowledge Base**
 - The system utilizes a **NoSQL Vector Database** to store high-dimensional numerical representations (embeddings) of Ethiopian legal documents.
 - By using semantic search rather than keyword matching, the system can understand the intent behind a user's question, even if they do not use exact legal terminology.
- **Decoupled Full-Stack Framework**
 - **Backend (Python Flask/ Javascript Node):** Acts as the orchestration layer, managing the logic for document ingestion, semantic search, and AI response generation.
 - **Frontend (Next.js):** Provides a modern, responsive interface designed for both mobile and desktop users. It ensures that complex legal information is presented through a simple, conversational chat experience.
- **Integrated Document & Identity Management**
 - **Better Auth & RBAC:** The solution incorporates a secure **Role-Based Access Control** system. This allows guest users to ask basic questions while providing administrators and legal experts with specialized tools to upload, edit, and audit the legal knowledge base.

- **Admin Dashboard:** A dedicated management interface allows for the continuous update of the vector store. As new proclamations are issued in the *Negarit Gazeta*, they can be ingested into the system instantly without requiring the AI model to be retrained.
- **Verifiable Citations and Transparency**
 - To build trust and ensure legal accuracy, every response generated by the chatbot is accompanied by **direct citations** to the specific article or proclamation number it was derived from. This allows users to cross-reference the AI's summary with the original legal text.

1.3 Objectives

The primary goal of this project is to bridge the gap between complex legal archives and the information needs of the Amharic-speaking public through the application of modern AI.

1.3.1 General Objective

The general objective of this project is to design, develop, and implement an **Amharic Legal Chatbot** based on **Retrieval-Augmented Generation (RAG)** architecture that provides accurate, accessible, and grounded legal information to users, thereby enhancing legal literacy and transparency in Ethiopia.

1.3.2 Specific Objectives

To achieve the general objective, the project focuses on the following technical and functional milestones:

- **Develop a Specialized Amharic Vector Store**
 - To digitize and preprocess unstructured Amharic legal PDFs (proclamations, regulations, and directives) and store them in a **ChromaDB** or MongoDB vector database using semantic embeddings for high-accuracy retrieval.
 - To implement a document chunking strategy that preserves the legal context of specific articles and sub-articles.
- **Build the RAG Orchestration Pipeline**
 - To integrate **Python Flask/ Javascript Node** and **LangChain** to create a pipeline that retrieves relevant legal context and augments Large Language Model (LLM) prompts to eliminate hallucinations and ensure factual grounding.
 - To design custom Amharic prompt templates that force the model to cite specific sources and acknowledge information gaps when data is unavailable.
- **Implement a Secure User Management System**

- To integrate **Better Auth** with **MongoDB** to provide a secure authentication layer and **Role-Based Access Control (RBAC)** for Guests, Registered Users, and Administrators.
- To develop a persistent chat history feature that allows users to track and revisit their previous legal inquiries.
- **Create an Administrative Knowledge Management Dashboard**
 - To build an interface for legal experts to upload new documents, manage vector embeddings, and update the knowledge base in real-time as new laws are ratified.
 - To implement comprehensive **Audit Logs** to monitor system interactions and ensure the integrity of the legal advice provided.
- **Design a Responsive Conversational Interface**
 - To develop a modern, user-friendly frontend using **Next.js** that supports natural Amharic text rendering and provides a seamless chat experience across both mobile and desktop devices.

1.4 Methodology

The methodology adopted for this project follows an iterative and structured approach to software development, ensuring that the **Amharic Legal Chatbot** is both technically robust and legally accurate. Given the complexity of Amharic linguistics and the high stakes of legal consultation, a hybrid of **Agile** and **Object-Oriented Analysis and Design (OOAD)** was selected to allow for continuous refinement of the RAG pipeline.

1.4.1 Requirement Gathering Methods

The success of a specialized AI system depends on the quality of its foundational requirements. To ensure the system meets the needs of both the public and legal experts, the following gathering methods were utilized:

- **Document Analysis (Primary Source Research)** This was the most critical phase. We conducted an exhaustive review of the **Federal Constitution of Ethiopia** to understand its hierarchical structure. This helped define the "metadata requirements" (e.g., date, proclamation number, and article hierarchy) needed to organize the vector database. While we prepared a comprehensive corpus of legal data gathered from the web—including *Negarit Gazeta* archives, legal blogs, and online Amharic law books—the current implementation focuses primarily on the Constitution. This scoping decision was made due to the technical constraints of available **free LLM APIs**, which impose strict **token limits** and offer limited support for specialized, low-resource languages like Amharic.
- **Informal Interviews** Semi-structured, informal interviews were conducted with **law students and teachers** in Ethiopia. These sessions aimed to identify potential "most

frequently asked questions" (FAQs) from citizens and pinpoint the specific legal jargon that currently acts as a barrier to public understanding. These insights guided the design of the conversational interface to ensure the chatbot uses language that is accessible yet legally grounded.

- **Technical Benchmarking:** We analyzed existing LLM limitations regarding the Amharic language. This gathering phase focused on identifying "hallucination patterns" in general-purpose AI when asked about Ethiopian law, which established the non-functional requirement for a strict "**groundedness**" policy in the AI responses.

1.4.2 Analysis and Design Methodology

For the system's structural blueprint, we employed **Object-Oriented Analysis and Design (OOAD)**. This methodology was chosen to manage the complexity of the RAG engine and the multi-layered architecture of the Next.js and Flask / **Javascript Node** stack.

- **Architectural Design (Decoupled Model):** The system was designed using a "Separation of Concerns" principle. The **Next.js** frontend is responsible for the View layer, **Flask/ Javascript Node** acts as the Controller/Logic layer, and **MongoDB/ChromaDB** serve as the Data layer. This ensures that a change in the AI model (the generator) does not require a rewrite of the entire user interface.
- **Vector Space and Embedding Design:** A critical part of the analysis involved determining how to "chunk" legal documents. We designed an **Article-Aware Chunking Strategy**, where each legal article is treated as a semi-independent object with its own metadata. This ensures that when the retriever finds a relevant paragraph, it brings the "parent" article context along with it.
- **Security and RBAC Design:** Using **Better Auth** as a framework, we designed a **Role-Based Access Control (RBAC)** model. This involved defining specific classes and middleware to handle "User," and "Admin" permissions, ensuring that only admins can modify the legal knowledge base.
- **Sequence Modeling:** We utilized sequence diagrams to map the flow of a query, from the moment an Amharic string is entered, through the vector search, into the prompt augmentation, and finally back to the user as a grounded response

1.4.3 Implementation Methodology

The implementation phase follows an **Iterative and Incremental (Agile)** approach. This allows for the "hill-climbing" of RAG quality, where the retrieval accuracy is improved through successive cycles of testing and refinement.

- **Sprints and Prototyping:** The development was divided into two-week sprints.
 - *Sprint 1:* Setup of the basic Flask or **Javascript Node** and Next.js environment and Better Auth integration.
 - *Sprint 2:* Implementation of the PDF ingestion pipeline and ChromaDB / MongoDB indexing.
 - *Sprint 3:* Development of the RAG logic and Amharic prompt engineering.
 - *Sprint 4:* UI polishing, history persistence, and admin dashboard completion.
- **The RAG Ingestion Pipeline:** Implementation began with the "Knowledge Layer." We built an automated pipeline using **LangChain** and **PyPDF** to clean Amharic text, remove OCR noise, and generate high-dimensional embeddings. These were then "upserted" into ChromaDB/MongoDB.
- **API Development and Integration:** We implemented the backend as a set of RESTful micro-services. The /ask endpoint was the focus of intensive unit testing to ensure that the AI remains within the bounds of the provided context.
- **CI/CD and Testing:** Continuous Integration (CI) was used to ensure that new code updates did not break existing authentication or chat history features. We performed "Groundedness Testing," where the AI's answers were manually compared against the original legal PDFs to verify that every citation was accurate.

1.5 Feasibility

This section evaluates whether the **Amharic Legal Chatbot** can be successfully developed and deployed within the current technical, financial, and time constraints of a final-year academic project.

1.5.1 Economic Feasibility

Since this is an academic research endeavor, the economic focus is on minimizing operational expenses while maximizing social value.

- **Development Costs:** The project utilizes an **Open-Source Tech Stack**. By leveraging **Python (Flask)/ Javascript (Node)**, **Next.js**, and **ChromaDB/MongoDB**, the initial development cost is effectively zero.
- **Operational Costs (API & Hosting):** To maintain the project's sustainability within a student budget, the implementation relies on **Free Tier LLM APIs** (such as Groq or limited OpenAI usage) and free hosting platforms like **Vercel** for the frontend and **Render** or **Railway** for the backend.
- **Long-term Value:** While the initial cost is low, the potential economic impact of democratizing legal knowledge in Ethiopia is immense. By reducing the need for physical travel to government offices and providing instant legal clarity, the system saves significant time and resources for the general public.

- **Hardware Requirements:** Development is performed on standard personal computing hardware, requiring no expensive specialized GPU clusters, as the RAG architecture offloads heavy computation to cloud-based LLM providers.

1.5.2 Technical Feasibility

This section assesses whether the current state of technology supports the project's goals, particularly regarding the Amharic language and the RAG architecture.

- **LLM Capabilities:** Modern Large Language Models have shown significant progress in understanding low-resource languages like Amharic. By using a **RAG pipeline**, we overcome the "knowledge gap" of these models, ensuring they do not need to "know" the law but only need the "linguistic ability" to summarize the Amharic text we provide.
- **Retrieval Infrastructure: ChromaDB or MongoDB** is a proven, lightweight vector database that can handle thousands of legal "chunks" with sub-second retrieval times. This ensures the system remains responsive even with a large legal corpus.
- **Integration Maturity:** The combination of **Next.js** for the frontend and **Flask/NodeJS** for the backend is a industry-standard architecture. Furthermore, the use of **Better Auth** provides a ready-made, secure solution for handling user sessions in MongoDB, reducing the technical risk of building security from scratch.
- **Data Availability:** Despite the fragmentation of legal documents, the **1995 Federal Constitution** and major proclamations are available in digital Amharic formats, making the creation of an initial knowledge base technically viable.

1.5.3 Time Feasibility

The project is designed to be completed within the standard academic semester (approximately 16 weeks).

- **Phased Development:** The project is broken down into manageable phases:
 - **Phase 1 (Weeks 1-4):** Requirement gathering, document collection, and architectural design.
 - **Phase 2 (Weeks 5-8):** Data preprocessing, vector store setup, and core RAG engine development.
 - **Phase 3 (Weeks 9-12):** Backend API development, Better Auth integration, and Frontend UI implementation.
 - **Phase 4 (Weeks 13-16):** Testing, refinement of Amharic prompts, and final documentation.

- **Scope Management:** To ensure completion on time, the project scope is carefully focused on the **Ethiopian Constitution** as the primary knowledge source. While the system is built to be scalable, this "Constitution-First" approach ensures a functional and valuable prototype is delivered by the deadline.
- **Rapid Prototyping:** Using **LangChain** significantly accelerates the development of the AI pipeline, allowing the team to focus on refining Amharic accuracy rather than building the underlying retrieval logic from scratch.

1.6 Beneficiaries or Significance of the Project

The development of the **Amharic Legal Chatbot** carries significant social, economic, and technical value for the Ethiopian legal ecosystem. By centralizing knowledge and simplifying access, the project provides unique benefits to three primary groups:

- **General Citizens**
 - **Democratic Access to Rights:** Citizens, especially those in rural areas, can learn about their constitutional rights and administrative procedures without the need for expensive legal consultants or long-distance travel to government offices.
 - **Simplification of "Legalese":** The system translates technical legal jargon into conversational Amharic, making complex laws understandable for non-experts.
 - **24/7 Availability:** Unlike physical law firms or public offices, the chatbot provides instant legal guidance at any time, reducing the frustration of waiting in queues for basic information.
- **Legal Professionals and Students**
 - **Enhanced Research Efficiency:** Law students and practitioners can use the tool to quickly locate specific articles within the Constitution or cross-reference proclamations, saving hours of manual document sifting.
 - **Decision Support:** By providing grounded, cited paragraphs, the system acts as a "research assistant" that helps professionals verify statutes before drafting legal opinions or case files.
 - **Focus on High-Value Tasks:** By automating the answering of "frequently asked questions," the system allows legal experts to focus on complex litigation and strategic advisory rather than repetitive information delivery.
- **Governmental and Administrative Institutions**
 - **Reduced Administrative Burden:** Public offices can integrate similar AI tools to handle basic inquiries, allowing civil servants to manage more complex administrative duties.

- **Promotion of Legal Literacy:** A more informed citizenry leads to higher compliance with the law and a stronger, more transparent relationship between the government and the people.

1.7 Limitations of the Project

While the system is a significant step toward legal accessibility, several technical and practical constraints define its current scope:

- **Constraint to Primary Knowledge (The Constitution)**
 - Due to the high cost of premium AI models and the **token limits** of free-tier LLM APIs, the current knowledge base is primarily focused on the **1995 Federal Constitution**. While a larger corpus (Negarit Gazeta, etc.) has been prepared, it is not yet fully integrated into the live chat to prevent performance degradation.
- **Linguistic Nuances and Translation Gaps**
 - Amharic is a morphologically rich and complex language. Current AI models, primarily trained on English datasets, may occasionally miss the subtle cultural or legal nuances of specific Amharic terms, leading to summaries that are accurate but less "natural" in tone.
- **Hardware and API Latency**
 - Since the project relies on cloud-based free-tier hosting (Vercel/Render) and third-party LLM APIs, users may experience slight delays (latency) in response generation. Furthermore, the system requires an active internet connection to function.
- **"Advice" vs. "Information"**
 - The system is designed to provide **legal information**, not **legal advice**. It cannot represent a user in court or provide strategic counseling for specific criminal or civil cases. It remains a tool for information retrieval and must be used as a supplement to professional legal counsel.
- **Dependency on Digital Source Quality**
 - The accuracy of the RAG engine is entirely dependent on the quality of the source PDFs. If the original legal document contains OCR (optical character recognition) errors or poor formatting, the AI may misinterpret the text.

1.8 Scope of the Project

The scope of this project defines the boundaries and the specific functional areas that the **Amharic Legal Chatbot** will cover. To ensure a high-quality and reliable prototype within the given timeframe, the scope is divided into the following domains:

- **Knowledge Domain:** * The primary source of legal truth for this prototype is the **1995 Federal Constitution of Ethiopia**.
 - While a broader dataset including *Negarit Gazeta* proclamations and legal blogs was prepared and chunked, the live RAG implementation is focused on the Constitution to maintain high precision and stay within the **token limits** of free-tier LLM APIs.
- **Target Language:** * The system is exclusively designed for the **Amharic language**. This includes processing Amharic PDF text, performing semantic searches in Amharic vector space, and generating natural language responses in Amharic.
- **Target Users:** * The project serves two specific roles: **Users** (nonpersistent and persistent chat history), and **Administrators** (managing the knowledge base).
- **Technical Implementation:** * Development of a full-stack application using **Next.js** (Frontend), **Flask/NodeJS** (Backend), **ChromaDB** / MongoDB (Vector Database), and **MongoDB** (User/Session Data).
 - The AI logic is limited to **Retrieval-Augmented Generation (RAG)**, focusing on information retrieval rather than complex legal reasoning or court representation.

1.9 Organization of the Project

Phase 1: Analysis & Requirements (Concept Phase)

This phase is the "Foundation." It defines the boundaries of what the system must do to solve the Ethiopian legal access problem.

- **Why it's useful:** It prevents "scope creep." By clearly defining the **Functional Features** (e.g., semantic search) and **Non-Functional Features** (e.g., zero-hallucination policy), it ensures the technical build in later phases stays focused on legal accuracy.
- **System Deliverable:** The Software Requirement Specification (SRS) for a legal RAG environment.

Phase 2: Architectural & Data Design (Blueprint Phase)

This phase is the "Brain." It translates the requirements into a technical map before any code is written.

- **Why it's useful:** It manages complexity. For a RAG system, the design of the **Vector Space** (how the Constitution is stored) is just as important as the UI. Using UML diagrams here ensures that the **Next.js frontend** and **Flask / NodeJS backend** communicate efficiently without data loss.

- **System Deliverable:** The System Architecture Diagram and Database Schema (ChromaDB + MongoDB).

Phase 3: Development & RAG Integration (Build Phase)

This phase is the "Body." It documents the actual coding and integration of the AI pipeline.

- **Why it's useful:** It provides the "Manual." It explains the specific implementation of the **RAG Pipeline**: how the system chunks Amharic text, how it creates embeddings, and how the prompt is "stuffed" with legal context to guide the LLM.
- **System Deliverable:** The functional API endpoints (e.g., /api/ask) and the integrated Amharic Chat Interface.

Phase 4: Evaluation & Quality Assurance (Validation Phase)

This phase is the "Verdict." It proves that the system actually works as a reliable legal tool.

- **Why it's useful:** It ensures safety. In legal tech, "it works" isn't enough; it must be accurate. This phase documents **Groundedness Testing**, where the system's output is compared against the original Constitution to ensure it never "invents" laws.
- **System Deliverable:** The Final Performance Report and Audit Logs.

Summary Table: Why Phase-Based Layout Works

Table 1: Phase based layout

Phase	System Value	Practical Benefit
Phase 1: Requirements	Problem Identification	Ensures the chatbot solves a <i>real</i> legal need.
Phase 2: Design	Structural Integrity	Prevents bugs by mapping the RAG logic early.
Phase 3: Implementation	Core Functionality	The actual "Construction" of the AI logic.
Phase 4: Evaluation	Reliability & Trust	Proves the Amharic law answers are 100% verified.

Chapter Two: System Features

2.1 The Existing System

The current landscape of legal information delivery in Ethiopia relies on traditional, manual, and paper-based processes. Despite recent efforts toward digitalization, the existing system remains fragmented, creating a "knowledge barrier" for the average citizen.

- **Manual and Fragmented Document Repositories**
 - Most legal documents (proclamations, regulations, and directives) are stored in decentralized silos. While some are available on ministry websites, they are often in non-searchable, static PDF formats or hosted on informal blogs.
 - Users must manually download and read through hundreds of pages to find a specific article, a process that is time-consuming and prone to human error.
- **High Dependency on Intermediaries**
 - Citizens currently rely heavily on legal professionals (lawyers and paralegals) for even the most basic legal clarifications. This creates a financial burden and limits legal access to those who can afford professional consultation.
 - In rural areas, physical distance to government offices and legal aid centers acts as a significant bottleneck.
- **Complexity of Legal Language (Legalese)**
 - The formal Amharic used in official gazettes is academic and highly specialized. There is no existing tool to "simplify" or "summarize" these complex texts into plain, conversational Amharic for the general public.
- **Inaccurate Informal Channels**
 - Without an official digital legal assistant, many citizens turn to unverified social media sources or general-purpose AI (like basic ChatGPT), which frequently **hallucinate** laws or cite foreign jurisdictions that do not apply to the Ethiopian legal context.

2.2 Proposed System

The proposed **Amharic Legal Chatbot** is a modern, AI-driven solution designed to democratize access to the Ethiopian Constitution and future legal codes. By utilizing a **Retrieval-Augmented Generation (RAG)** architecture, the system replaces manual searching with an intuitive, conversational experience.

- **RAG-Based Grounding and Accuracy**
 - Unlike standard AI, the proposed system is "grounded." Before answering, it searches a private, verified database (the **Vector Store**) containing the 1995

Ethiopian Constitution. It then uses only that specific text to generate an answer, virtually eliminating the risk of misinformation.

- **Natural Language Amharic Interface**
 - The system allows users to ask questions in everyday Amharic. The backend uses semantic search to understand the user's intent, even if they do not use exact legal terminology (e.g., searching for "human rights" will correctly retrieve "Fundamental Rights and Freedoms").
- **Automated Source Attribution**
 - Transparency is a core feature. For every answer provided, the system automatically cites the specific **Article number** it referenced. This allows users to cross-check the AI's summary against the official legal text.
- **Persistent History and User Management**
 - Through a secure **Next.js and MongoDB** integration, the system offers a personalized "User Workspace." Registered users can save their legal inquiries and return to them later, facilitating complex, multi-step legal research.
- **Scalable Knowledge Management**
 - The system includes an **Admin Dashboard** where legal experts can upload new proclamations. The system then "chunks" and "indexes" these new laws into **ChromaDB / MongoDB** instantly, ensuring the chatbot stays up-to-date without needing complex technical re-training.

2.3 Requirement Analysis

2.3.1 Functional Requirements

Login: Users and Administrators log into the system using their secure credentials via the **Better Auth** interface to access personalized chat history and administrative tools.

Logout: Users and Administrators log out of the web application to terminate their session and secure their data.

Create Account: The system allows new users to register an account, which is stored in **MongoDB** to enable the saving of legal inquiry threads.

Submit Amharic Query: Users submit legal questions in natural Amharic text through the chat interface to initiate the AI retrieval process.

Search Legal Context: The system performs a semantic search within **ChromaDB / MongoDB** to identify and retrieve the most relevant sections from the uploaded legal PDF documents.

Generate Legal Answer: The system utilizes the RAG engine to synthesize a grounded Amharic response, ensuring the answer is based strictly on the retrieved legal text.

View Citations: The system displays the specific **Page Numbers** from the source PDF alongside the AI's response, allowing users to verify the information in the original document.

Manage Chat History: Users can view, rename, and delete their previous legal inquiry threads, which are retrieved from the **MongoDB** database.

Upload Legal Documents: Administrators upload new legal PDF documents (Proclamations or Gazettes) to the system to expand the AI's knowledge base.

Update Vector Embeddings: The system automatically processes newly uploaded PDFs, converting the text into numerical vectors and storing them in **ChromaDB/MongoDB** for immediate retrieval.

Manage Knowledge Base: Administrators can view, update, or delete indexed legal documents to ensure the system's information remains accurate and authorized.

Manage User Accounts: Administrators can add, modify, or delete user accounts to maintain system security and oversight.

Provide Feedback: Users can rate the accuracy of the legal information provided, allowing for system monitoring and quality improvement.

Generate Audit Reports: The system generates reports on query trends and system usage for administrative review and legal research analysis.

System Updating: The system allows for the future transition of the citation engine from **Page Numbers** to specific **Legal Articles** as the data preprocessing becomes more granular.

Error Notification: The system displays user-friendly Amharic error messages if the AI service is unavailable or if the requested information is outside the scope of the database.

2.3.2 System Use Case

2.3.2.1 Use Case Diagram

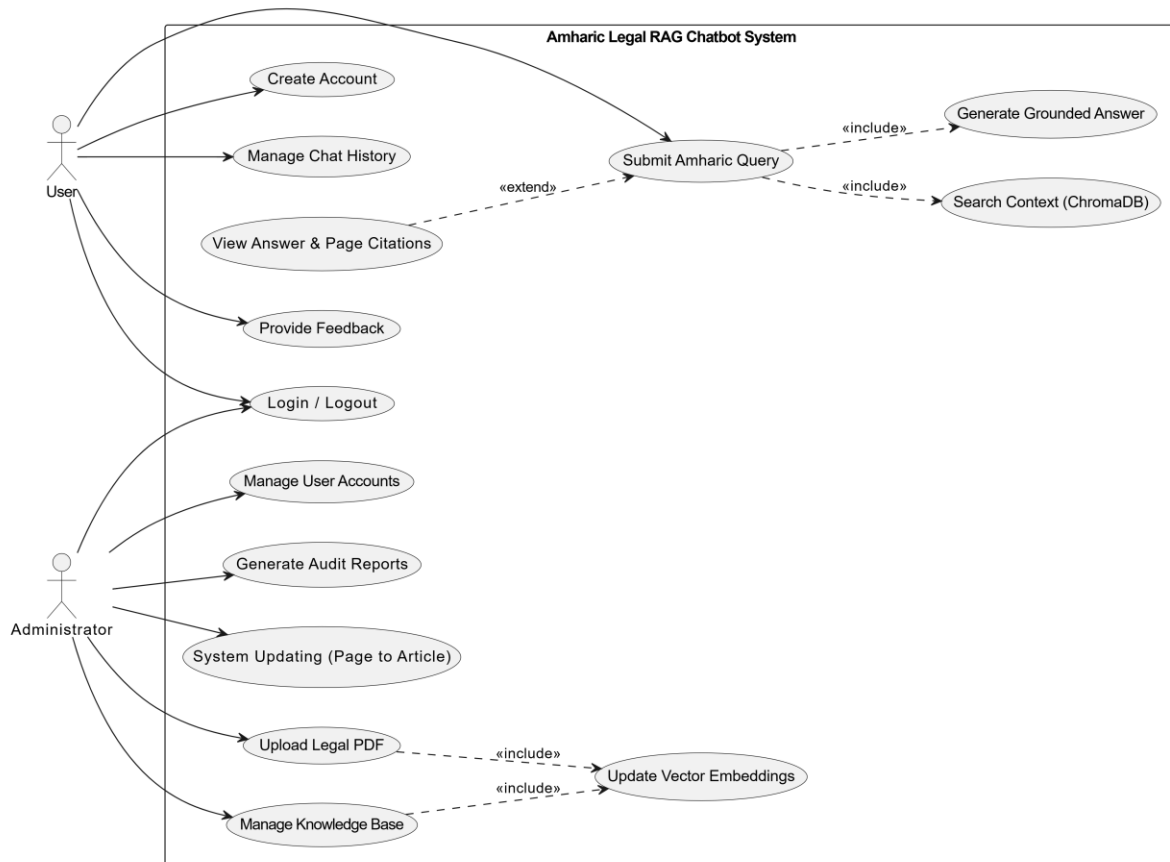


Figure 1: use case diagram

2.3.2.2 Use Case Documentation

User-Facing Use Cases

Table 2: UC-01

Use Case Detail	UC 01: Create Account
Priority	High
Actor	User
Description	Allows visitors to register by providing details to become recognized users.
Pre-condition	Visitor must have a valid email and access to registration page.
Post-condition	User account created and stored in MongoDB.
Extend	_____
Include	_____
Basic Action	1. Visitor clicks “Sign Up”.

	2. System displays registration form. 3. Visitor enters name, email, and password. 4. System validates and hashes password. 5. System saves record to database. 6. Use case end.
Alternative	If email already exists, system displays "Email already registered."

Table 3:UC-02

Use Case Detail	UC 02: Login / Logout
Priority	High
Actor	User / Administrator
Description	Allows actors to access their accounts or terminate sessions securely.
Pre-condition	Actor must have a registered account in MongoDB.
Post-condition	Session started (Login) or destroyed (Logout).
Extend	_____
Include	_____
Basic Action	1. Actor enters email and password. 2. System validates via Better Auth . 3. System redirects to dashboard. 4. For logout, actor clicks “Logout”. 5. System terminates session. 6. Use case end.
Alternative	If credentials are wrong, system displays "Invalid email or password."

Table 4:UC-03

Use Case Detail	UC 03: Submit Amharic Query
Priority	High
Actor	User
Description	Allows users to ask legal questions in Amharic and receive AI

	answers.
Pre-condition	User is on the chat page; ChromaDB/MongoDB must be connected.
Post-condition	Legal answer generated and displayed.
Extend	UC 04 (View Page Citations)
Include	UC 02 (Login), UC 13 , UC 14
Basic Action	1. User types question in Amharic. 2. System converts text to vector embedding. 3. System searches ChromaDB/ MongoDB for legal context. 4. System sends context to AI model. 5. System displays answer. 6. Use case end.
Alternative	If no text found, system displays "No legal info found."

Table 5:UC-04

Use Case Detail	UC 04: View Answer & Page Citations
Priority	High
Actor	User
Description	Lets the user see the exact page number of the source PDF.
Pre-condition	A successful query must have been processed.
Post-condition	Source page number displayed to user.
Extend	_____
Include	UC 03 (Submit Query)
Basic Action	1. System identifies metadata of retrieved chunk. 2. System extracts "Page Number" field. 3. System appends page number to AI response. 4. User views the citation. 5. Use case end.
Alternative	If metadata is missing, system displays "Source unavailable."

Table 6:UC-05

Use Case Detail	UC 05: Manage Chat History
Priority	Medium

Actor	User
Description	Lets users view, rename, or delete past legal conversations.
Pre-condition	User must be logged in.
Post-condition	History updated in MongoDB.
Extend	_____
Include	UC 02 (Login)
Basic Action	1. User opens "History" sidebar. 2. System fetches threads from MongoDB. 3. User selects a thread to read or delete. 4. System updates UI or removes record. 5. Use case end.
Alternative	If no history, system displays "No previous chats."

Table 7:UC-06

Use Case Detail	UC 06: Provide Feedback
Priority	Low
Actor	User
Description	Lets the user rate the accuracy of the AI-generated answer.
Pre-condition	User must have received an answer.
Post-condition	Feedback stored for Admin review.
Extend	_____
Include	UC 02 (Login)
Basic Action	1. User clicks "Like" or "Dislike" on a message. 2. System records rating and message ID. 3. System saves feedback to MongoDB. 4. Use case end.
Alternative	_____

Administrator Use Cases

Table 8:UC-07

Use Case Detail	UC 07: Upload Legal PDF
Priority	High
Actor	Administrator

Description	Lets the admin add new legal documents to the knowledge base.
Pre-condition	Admin must be logged in with admin privileges.
Post-condition	PDF processed and ready for indexing.
Extend	_____
Include	UC 02, UC 08
Basic Action	1. Admin navigates to "Upload" section. 2. Admin selects a legal PDF. 3. System parses Amharic text. 4. System triggers indexing. 5. Use case end.
Alternative	If file is not PDF, system displays "Invalid format."

Table 9:UC-08

Use Case Detail	UC 08: Update Vector Embeddings
Priority	High
Actor	Administrator (System Triggered)
Description	Converts PDF text into vectors and stores them in ChromaDB / MongoDB.
Pre-condition	A PDF must be successfully uploaded.
Post-condition	ChromaDB / MongoDB updated with new searchable vectors.
Extend	_____
Include	_____
Basic Action	1. System splits text into semantic chunks. 2. System sends chunks to embedding model. 3. System saves vectors and page tags to ChromaDB / MongoDB. 4. System confirms success to Admin. 5. Use case end.
Alternative	If indexing fails, system displays "Error in processing."

Table 10:UC-09

Use Case Detail	UC 09: Manage Knowledge Base
Priority	Medium

Actor	Administrator
Description	Lets admin view or delete indexed legal documents.
Pre-condition	Admin must be logged in.
Post-condition	Knowledge base records updated.
Extend	_____
Include	UC 02 (Login)
Basic Action	1. Admin clicks "Manage Documents". 2. System displays list of indexed PDFs. 3. Admin selects document to delete. 4. System removes vectors from ChromaDB/MongoDB. 5. Use case end.
Alternative	If no documents, system displays "Knowledge base empty."

Table 11:UC-10

Use Case Detail	UC 10: Manage User Accounts
Priority	Medium
Actor	Administrator
Description	Lets the admin monitor or delete user accounts.
Pre-condition	Admin must be logged in.
Post-condition	User records updated in MongoDB.
Extend	_____
Include	UC 02 (Login)
Basic Action	1. Admin views "Users" list. 2. System displays emails and registration dates. 3. Admin selects user to deactivate or delete. 4. System updates MongoDB. 5. Use case end.
Alternative	If user not found, system displays "Error fetching user."

Table 12:UC-11

Use Case Detail	UC 11: System Update (Page to Article)
Priority	Low
Actor	Administrator

Description	Updates citation logic from page numbers to specific articles.
Pre-condition	Updated parsing scripts must be available.
Post-condition	ChromaDB / MongoDB metadata updated to article-level.
Extend	_____
Include	UC 02 (Login)
Basic Action	1. Admin initiates "Deep Indexing" script. 2. System re-scans PDFs for Article headers. 3. System replaces Page tags with Article tags. 4. Use case end.
Alternative	If parsing fails, system reverts to Page tags.

Table 13:UC-12

Use Case Detail	UC 12: Generate Audit Reports
Priority	Medium
Actor	Administrator
Description	Shows statistics on user queries and system usage.
Pre-condition	Admin must be logged in.
Post-condition	Report generated and displayed.
Extend	_____
Include	UC 02 (Login)
Basic Action	1. Admin clicks "View Reports". 2. System aggregates logs from MongoDB. 3. System displays charts of common topics. 4. Use case end.
Alternative	If no logs, system displays "Insufficient data."

Internal System Use Cases

Table 14:UC-13

Use Case Detail	UC 13: Search Context
Priority	High
Actor	System (Internal)
Description	Retrieves relevant text segments from the vector database.
Pre-condition	Query must be vectorized and ChromaDB / MongoDB must be online.

Post-condition	Top-\$k\$ relevant text chunks retrieved.
Extend	_____
Include	_____
Basic Action	1. System receives query vector. 2. System calculates similarity in ChromaDB/MongoDB. 3. System extracts top matches. 4. System passes chunks to the generation module. 5. Use case end.
Alternative	If DB is offline, system returns a database error.

Table 15:UC-14

Use Case Detail	UC 14: Generate Grounded Answer
Priority	High
Actor	System (Internal)
Description	Uses an LLM to create a response based on retrieved data.
Pre-condition	Relevant text context must be successfully retrieved.
Post-condition	Response generated for user delivery.
Extend	_____
Include	_____
Basic Action	1. System formats retrieved text into a prompt. 2. System sends prompt to LLM. 3. System receives Amharic text response. 4. System passes response for display. 5. Use case end.
Alternative	If LLM fails, system returns a generation error.

2.3.3 Business Rule Documentation

Business rules define the constraints, operations, and logic that govern how the **Amharic Legal RAG Chatbot** functions. For an examiner, these rules explain the "policy" behind the code, ensuring the system remains legally reliable and secure.

Accuracy and Groundedness Rules (RAG Logic)

These rules ensure the system provides factual legal information rather than "hallucinating" or making up laws.

- **BR-01: Source Grounding:** The system must generate responses based *only* on the text segments retrieved from the uploaded legal documents. If the information is not present in the database, the system must state that it cannot find the answer rather than guessing.
- **BR-02: Mandatory Citation:** Every AI-generated response must be accompanied by a citation (initially a Page Number) to the original document.
- **BR-03: Language Integrity:** All user-facing legal summaries and system responses must be delivered in the **Amharic** language to maintain accessibility for the target demographic.

Access and Security Rules (Authentication)

These rules govern who can use the system and what actions they are permitted to perform.

- **BR-04: Session Timeout:** For security, user sessions managed by **Better Auth** must expire after a period of inactivity to prevent unauthorized access on shared devices.
- **BR-05: Admin Exclusivity:** Only users with the "Administrator" role are permitted to access the /admin portal, upload PDF files, or modify the Vector Knowledge Base.
- **BR-06: Account Uniqueness:** Every user must be registered with a unique email address to prevent duplicate profiles in the **MongoDB** database.

Knowledge Management Rules (Data Governance)

These rules define how legal data is handled and updated.

- **BR-07: File Integrity:** The system shall only be given and/or accept legal documents in **PDF format** to ensure that text parsing and page-numbering remain consistent for citations.
- **BR-08: Incremental Indexing:** When a new document is uploaded, it must be "chunked" and "embedded" before it is added to **ChromaDB / MongoDB**. Existing documents must not be deleted or overwritten unless explicitly commanded by an Administrator.

- **BR-09: Version Control:** If a law is repealed or updated by a new proclamation, the Administrator must mark the old document as "Archived" or remove it to ensure the chatbot only provides the most current legal advice.

2.3.4 User Interface Prototype

The UI is designed using a **Mobile-First** approach, ensuring that legal information is accessible on various devices. The layout uses a minimalist "Legal Professional" aesthetic with a focus on readability.

I. User Chat Interface (The "Legal Assistant")

This is the primary screen for the Standard User. It mimics a modern messaging application to reduce the learning curve.

- **Header:** Contains the application name and a sidebar toggle for **Chat History**.
- **Message Area:**
 - * **User Bubbles:** Right-aligned, showing the Amharic query.
 - **AI Bubbles:** Left-aligned, featuring a distinct icon. Answers are formatted in clear paragraphs.
- **Citation Box:** A small, shaded box at the bottom of the AI response labeled "Source" (e.g., *Source: 1995 Constitution, Page 12*).
- **Input Area:** A fixed bottom bar with a text input field and a "Send" icon.

II. Administrator Knowledge Dashboard

This interface is restricted to users with elevated privileges. It focuses on data management and system health.

- **Document Management Table:** Displays a list of all indexed PDFs, their upload date, and a "Delete" button to remove outdated laws from ChromaDB / MongoDB.
- **Upload Widget:** A "Drag and Drop" area that accepts PDF files. Once a file is dropped, a **Progress Bar** indicates the status of the "Text Parsing" and "Vector Embedding" stages.
- **User Management:** A simple list of registered users with the ability to manage roles or deactivate accounts.

III. Authentication & Onboarding

- **Splash Screen:** Features the logo and a "Get Started" button.
- **Login/Signup Form:** A clean, centered card with fields for Email and Password, powered by **Better Auth**.

Design Justification

1. **Transparency:** By showing the citation on the main chat screen, the UI reinforces **BR-02** (Mandatory Citation), building trust with the user.
2. **Simplicity:** The Admin dashboard provides visual feedback during the "Vector Embedding" process (UC 08), making a complex technical task easy to monitor.
3. **Privacy:** User history is tucked away in a sidebar, keeping the focus on the current legal inquiry while ensuring **BR-10** (History Privacy) is maintained.



Figure 2: CHAT UI

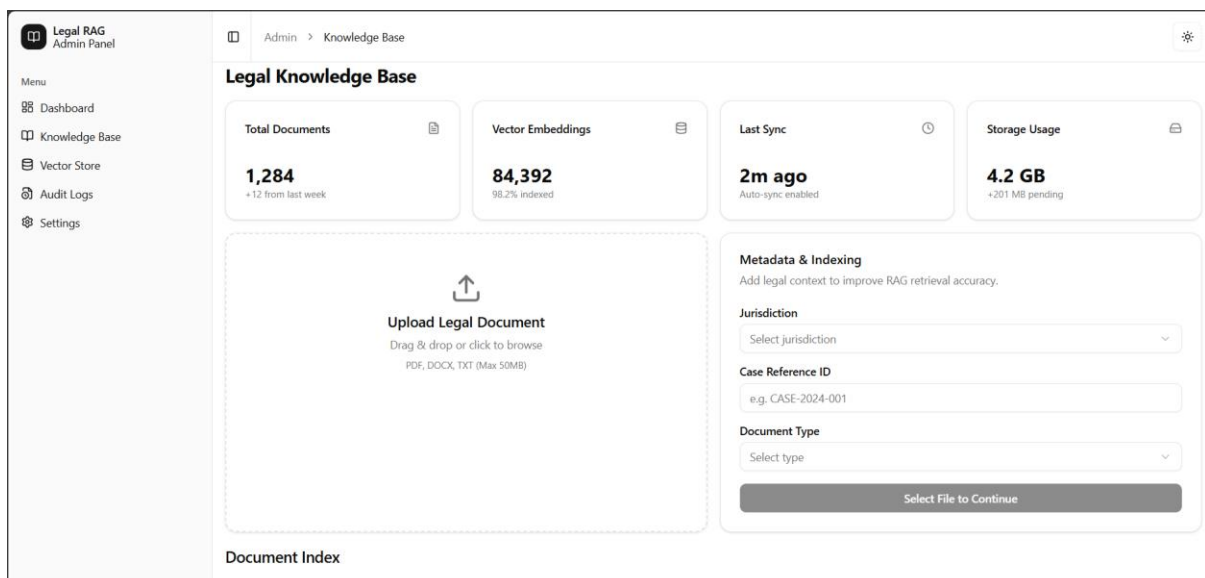


Figure 3:Admin Upload



Figure 4:Admin Dashboard

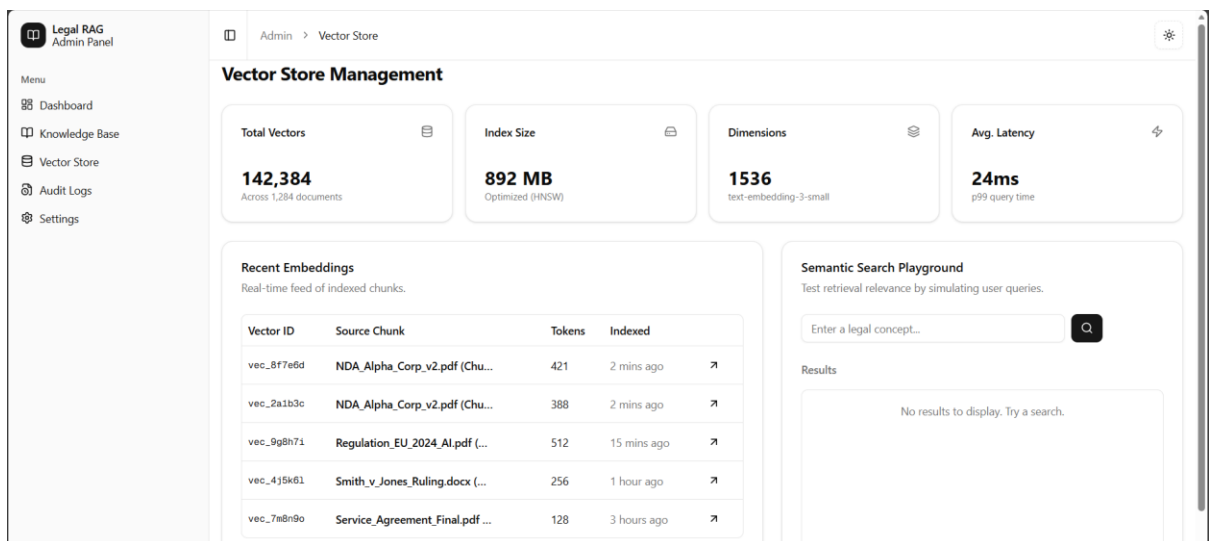


Figure 5: Vector store

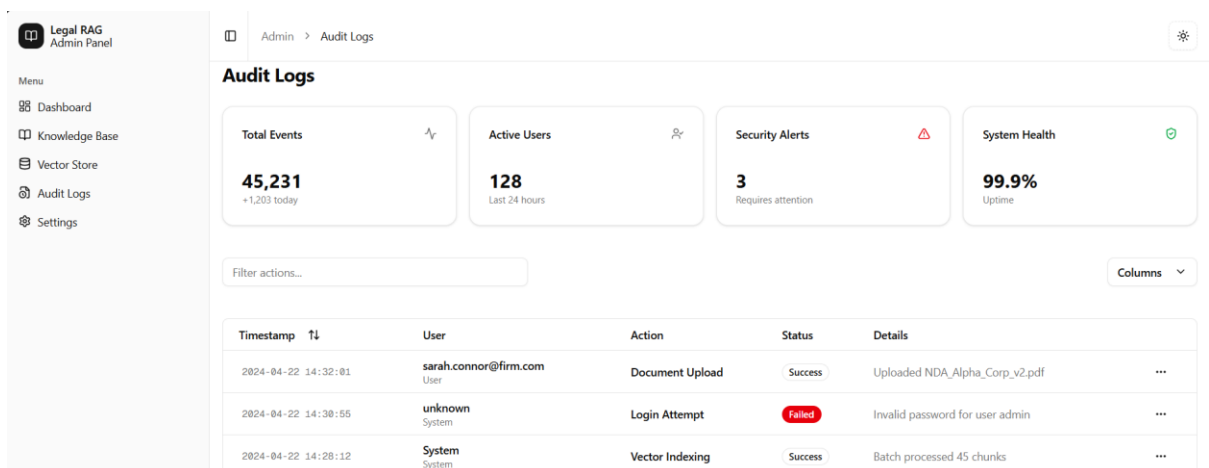


Figure 6: Admin Audit Lods

2.3.5 State Chart Diagram

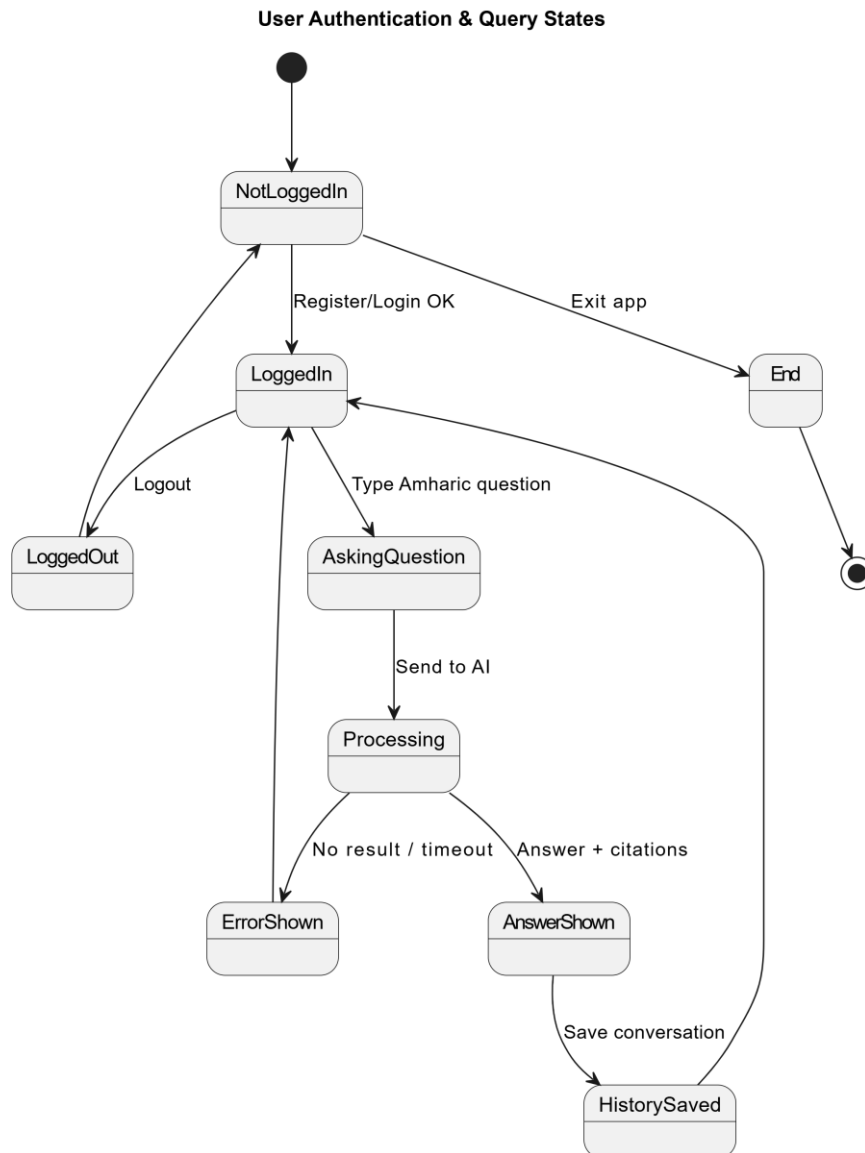


Figure 7: user auth Statechart diagram

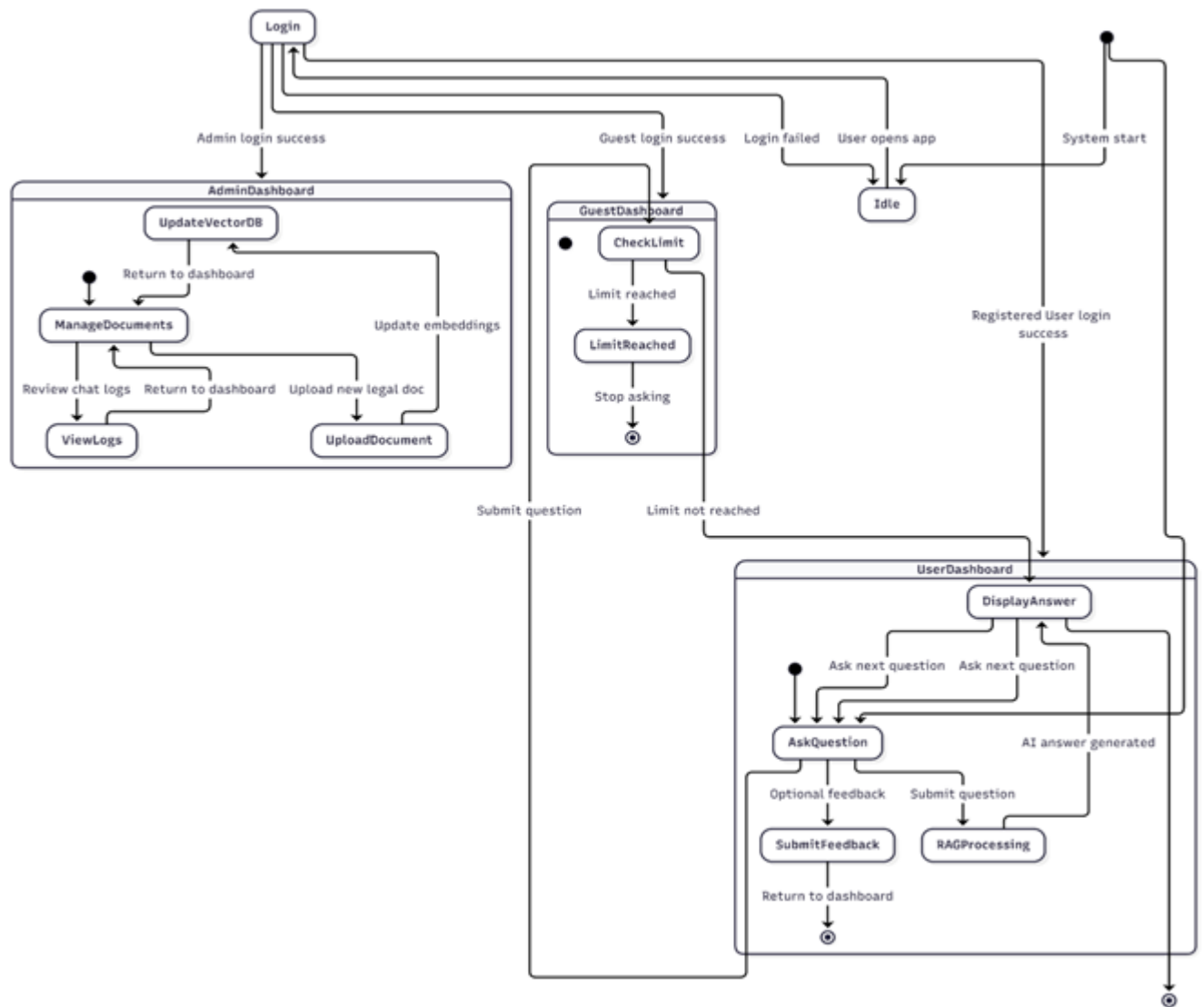


Figure 8: full system Statechart diagram

2.3.6 Activity Diagram

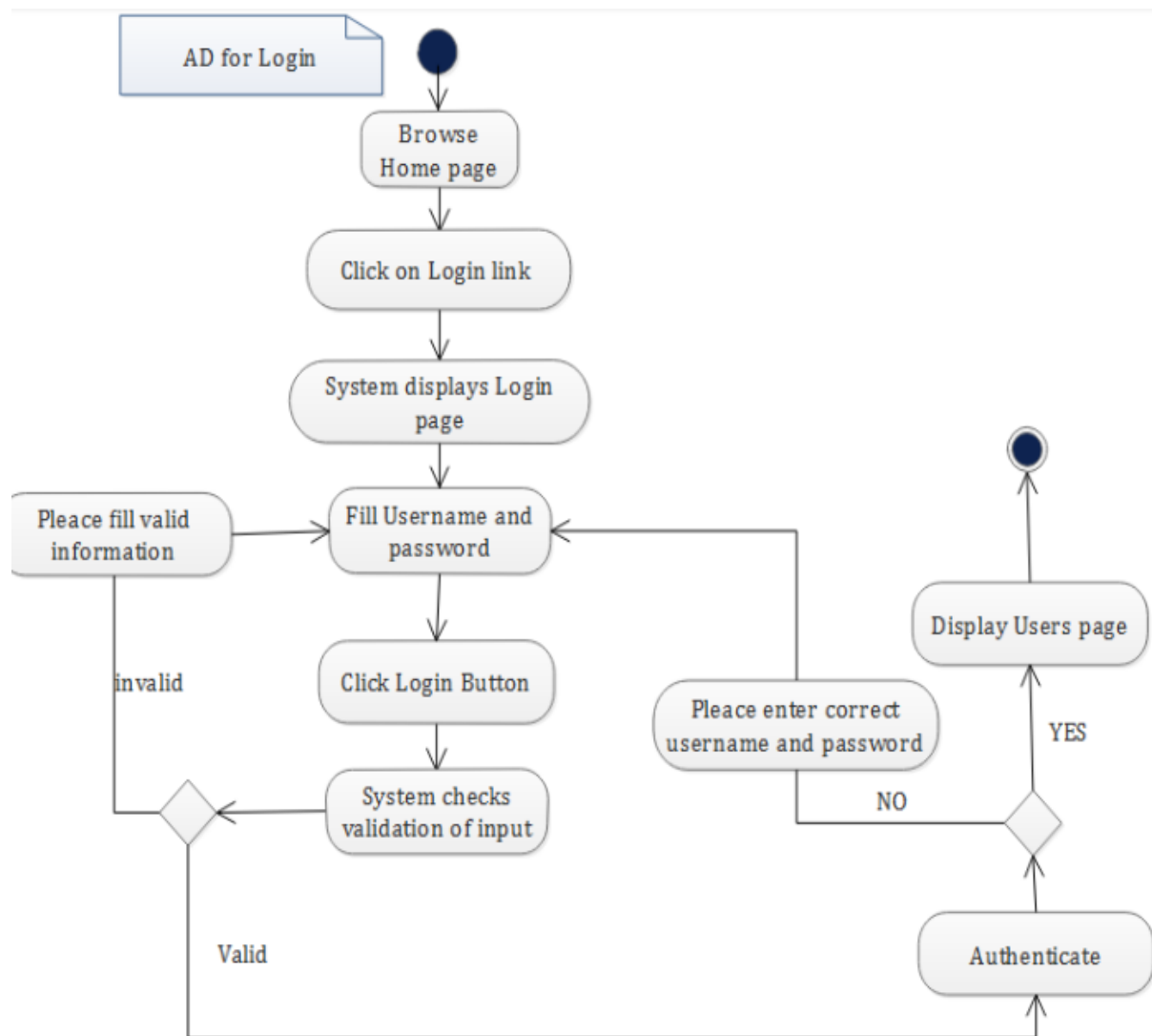


Figure 9: user authentication Activity Diagram

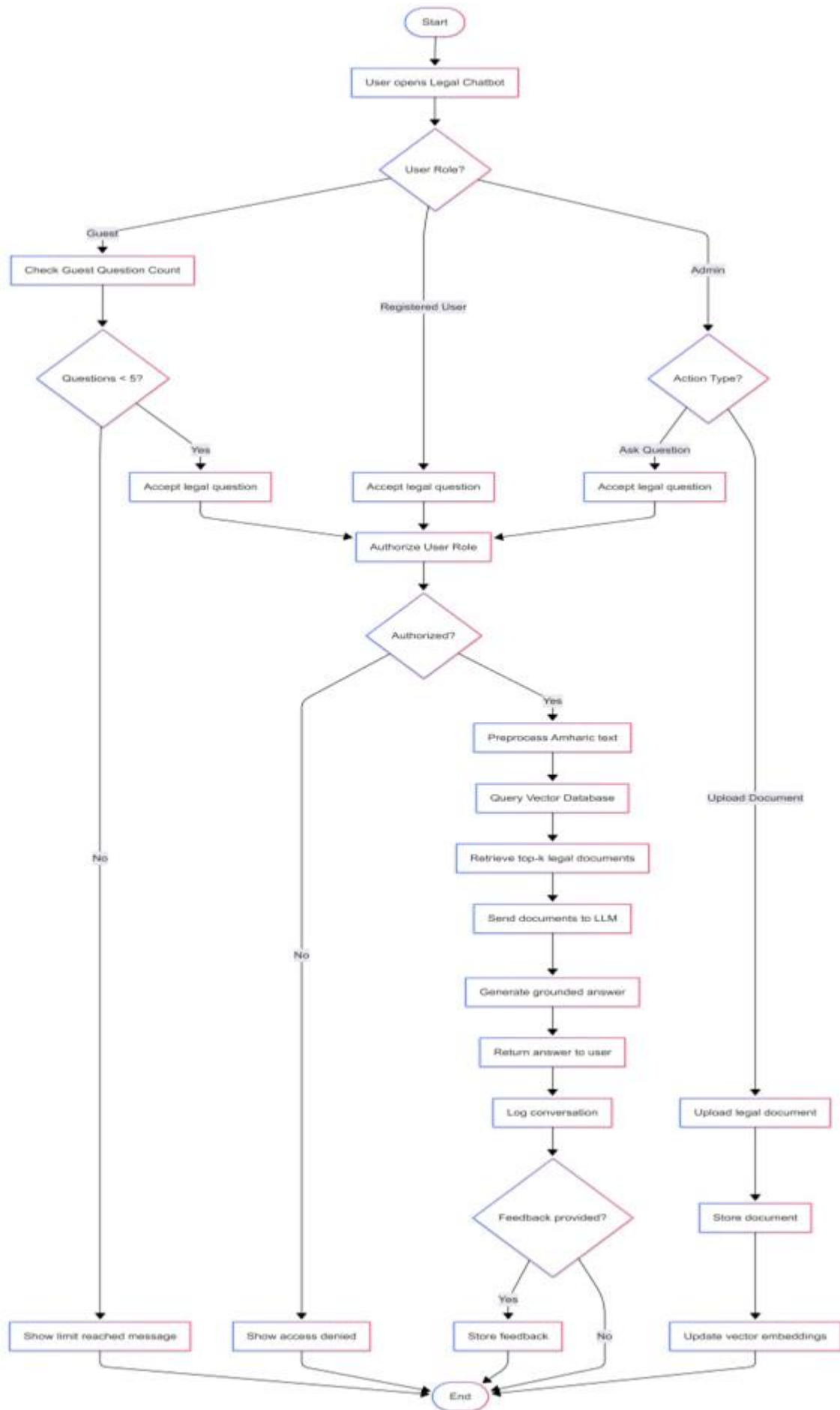


Figure 10: full system Activity Diagram

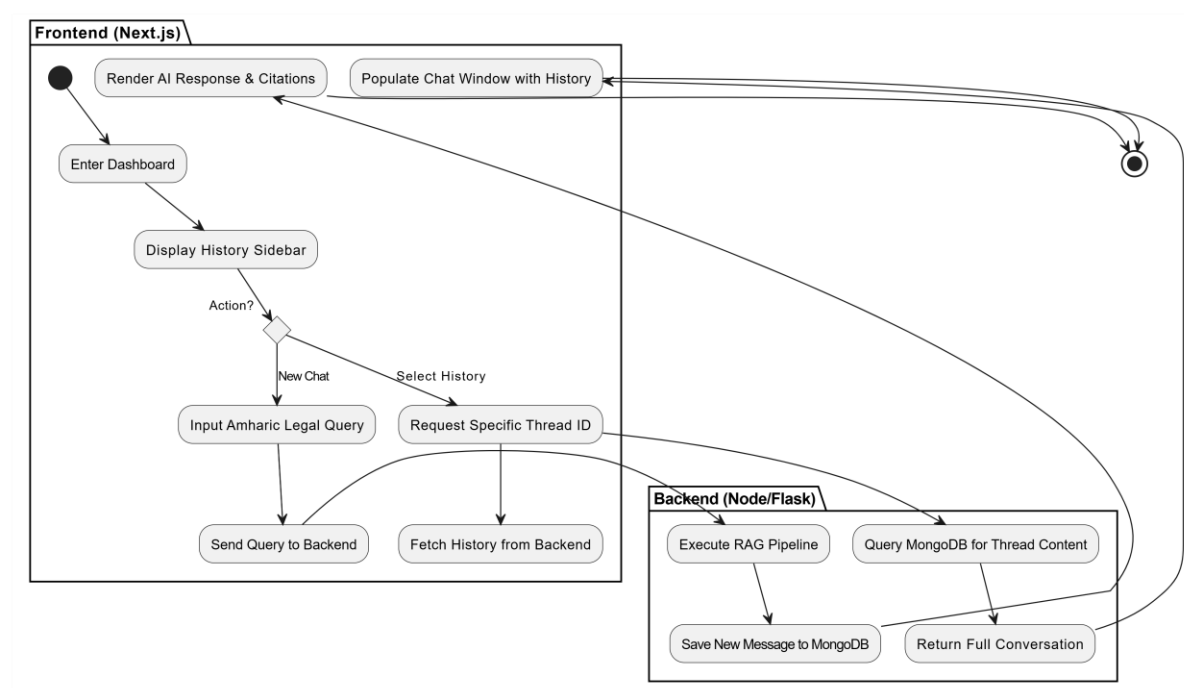


Figure 11: Show History Activity diagram

2.3.7 Sequence Diagram

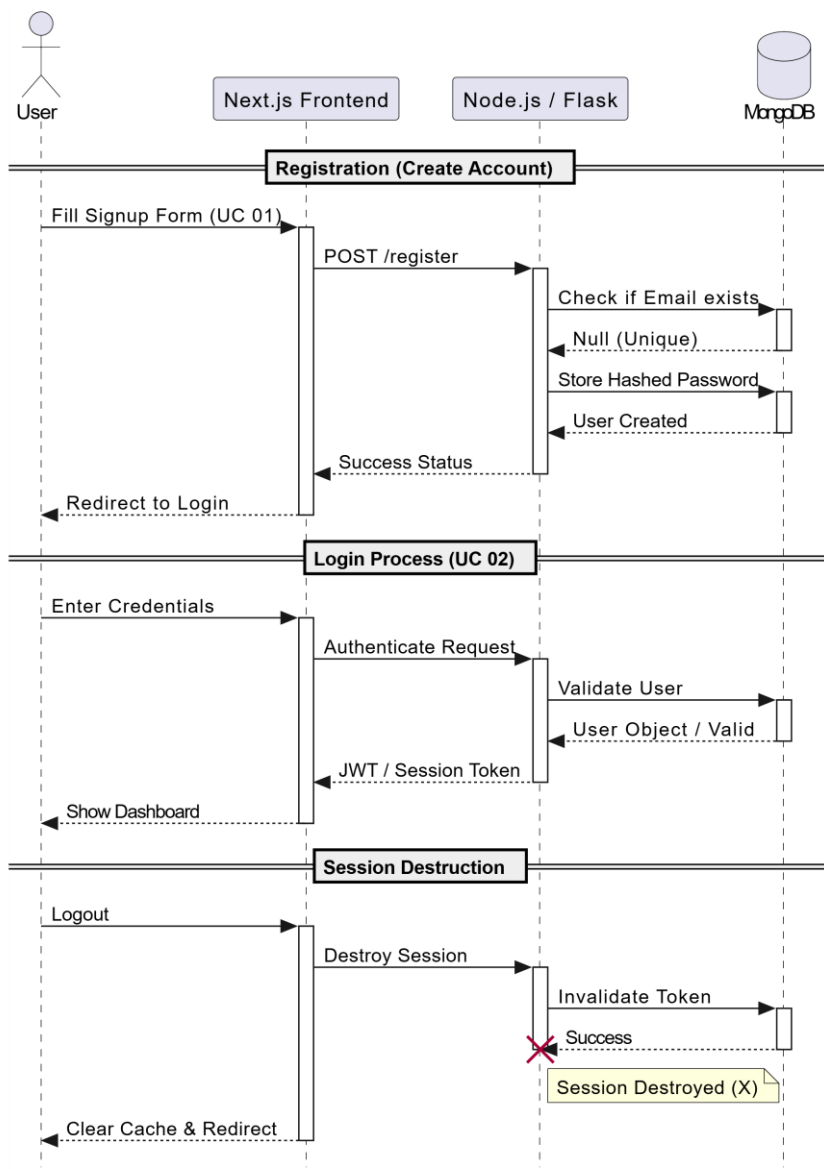


Figure 12 User Auth Sequence Diagram

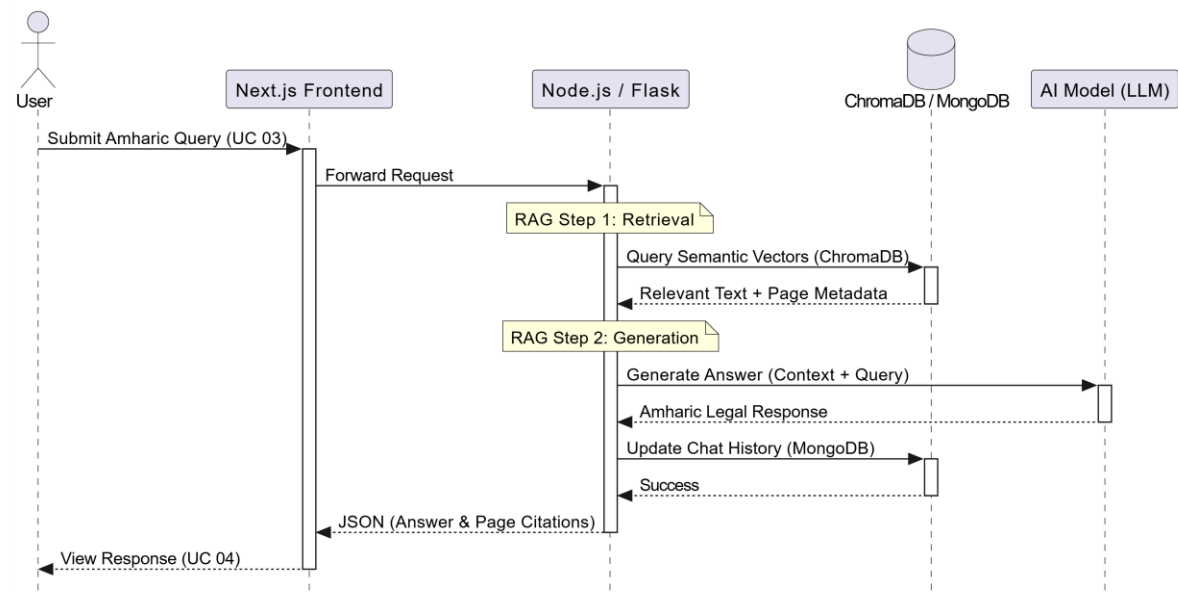


Figure 13: User Chatbot Interaction SequenceDiagram

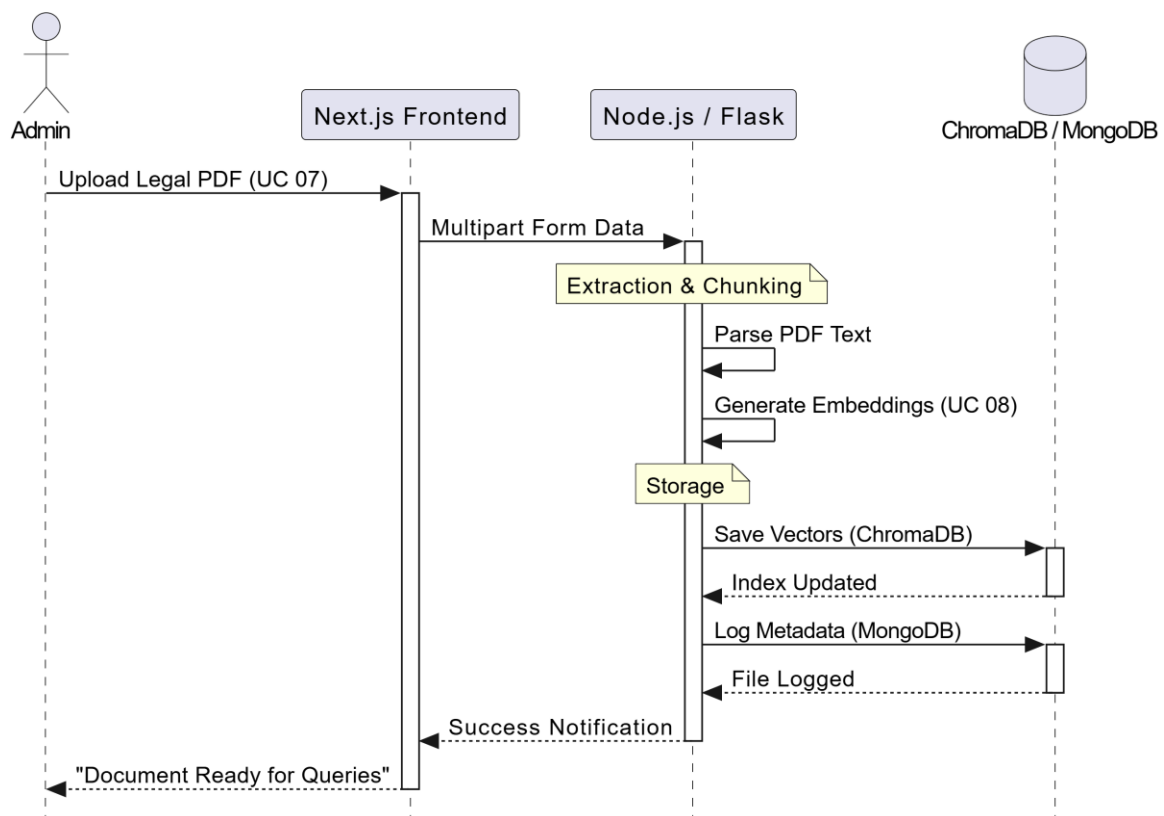


Figure 14: Admin Upload sequence Diagram

2.3.8 Analysis Class Model

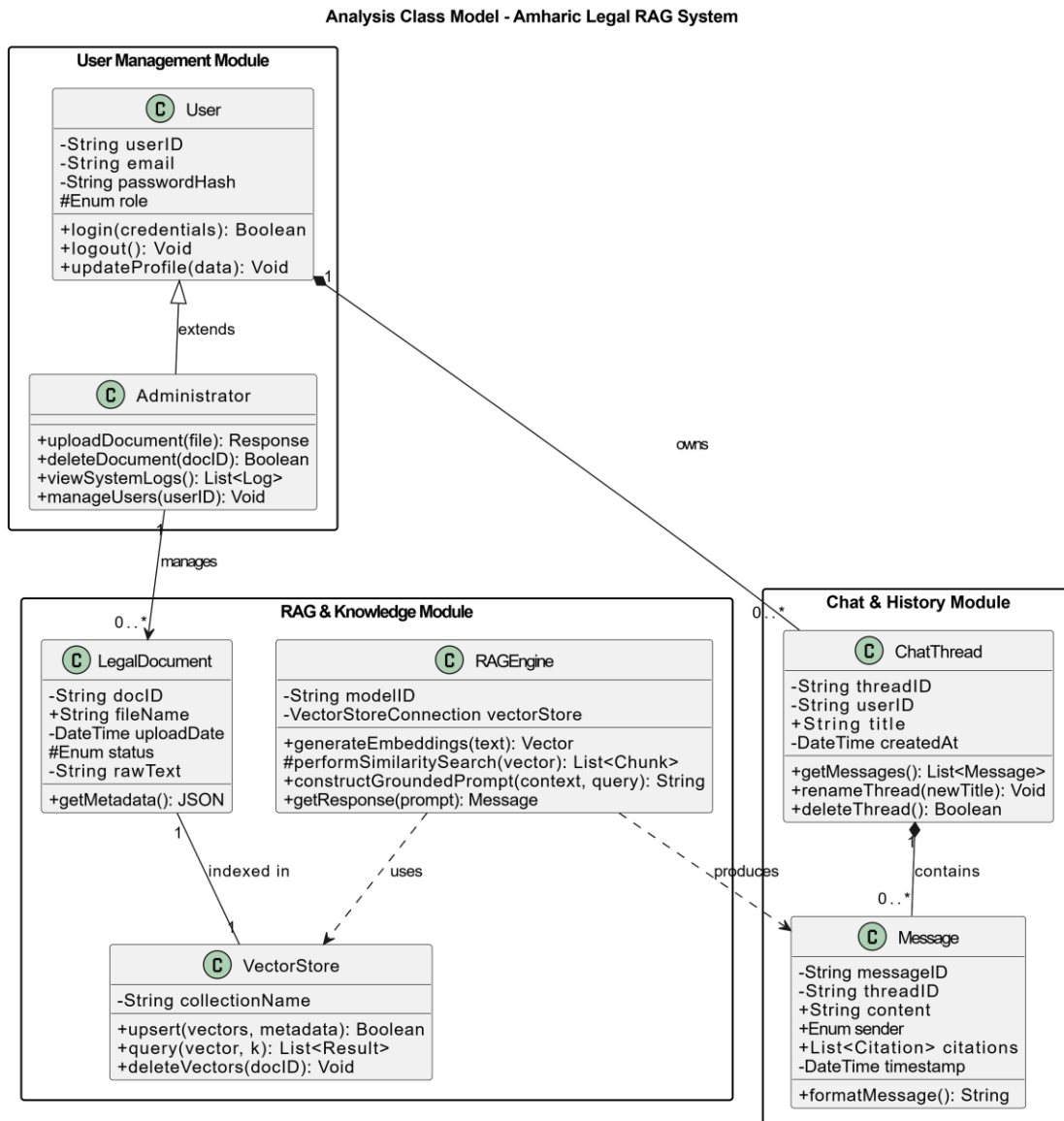


Figure 15: Analysis Class Model

2.3.9 Logic Model (Algorithm/Pseudo-code)

I. Algorithm for User Authentication & RBAC

This logic ensures that users are identified and their permissions are strictly enforced.

1. **Input:** Capture email and plain_text_password from the Next.js login form.
2. **Request:** Send credentials to the Node.js backend via a secure HTTPS POST request.
3. **Verification:** * Retrieve the stored password_hash from **MongoDB** based on the email.
 - a. Compare the hash with the input password.

4. **Decision:**
 - a. **IF** match fails: Return Unauthorized Error and log the failed attempt.
 - b. **ELSE:** Proceed to Step 5.
5. **Role Retrieval:** Fetch the role attribute (User or Admin) from the user's database document.
6. **Session Initiation:** Generate a JWT or **Better Auth** session containing the userID and role.
7. **Termination:** Redirect the user to their specific dashboard (Chat for Users, Management for Admins). **(X)**

II. Algorithm for the RAG Pipeline Processing

This is the core "intelligence" loop for processing Amharic legal queries.

1. **Input:** Receive the Amharic query string from the Frontend.
2. **Pre-processing:** Normalize Amharic characters (handling variations in Ethiopic script).
 - a. Remove stop-words and punctuation.
3. **Vectorization:** Pass the cleaned text to the embedding model to generate a numerical vector.
4. **Semantic Retrieval:** Query **ChromaDB** using the vector.
 - a. Retrieve the top-k (e.g., 3 to 5) relevant legal chunks based on cosine similarity.
5. **Context Augmentation:** Combine the retrieved legal text, page metadata, and the original query into a "System Prompt."
6. **Inference:** Send the augmented prompt to the **Gemini Pro LLM**.
7. **Output:** Receive the grounded Amharic response and return it to the backend controller. **(X)**

III. Algorithm for Response Delivery & Logging

This ensures the user sees the answer and the history is preserved for the sidebar.

1. **Reception:** Backend receives the generated response and source citations from the RAG engine.
2. **Delivery:** Send a JSON object containing the answer and citations to the Next.js Frontend.
3. **Rendering:** Frontend updates the state to display the message in the chat window.
4. **Persistence:** Create a new entry in the **MongoDB** Messages collection.
 - a. Link the message to the current ThreadID.
5. **Optional Feedback:** * **IF** the user clicks "Thumbs Up/Down": Capture the boolean and optional comment.

- a. **STORE** feedback in the Feedback collection for model fine-tuning. **(X)**

IV. Algorithm for Admin Document Ingestion

This covers how the Knowledge Base is updated with new Amharic laws.

1. **Upload:** Admin selects a PDF and submits it through the Admin Dashboard.
2. **Authorization Check:** Node.js verifies the session token has role: "Admin".
3. **Extraction:** * Flask backend uses a PDF parser (or OCR if scanned) to extract raw Amharic text.
4. **Chunking:** The text is split into smaller, overlapping segments (e.g., 500 characters) to preserve context.
5. **Indexing:**
 - a. **FOR EACH** chunk: Generate an embedding and save it to **ChromaDB**.
 - b. **SAVE** the original file metadata (file name, date, total chunks) to **MongoDB**.
6. **Notification:** Send a "Success" signal to the UI to update the document list.

2.4 Non-Functional Requirements

2.4.1 Performance and Scalability

- **Response Latency:** The RAG pipeline must generate a grounded legal answer within **5 to 8 seconds** from the time of query submission to ensure a fluid user experience.
- **Concurrent Users:** The backend (Node.js/Flask) should handle up to **50 concurrent sessions** without a significant degradation in retrieval speed.
- **Scalability:** The system must use a vector database (**ChromaDB**) capable of indexing thousands of legal articles while maintaining $O(n \log n)$ search efficiency.

2.4.2 Reliability and Availability

- **Uptime:** The system aims for **99.5% availability** during business hours to provide consistent access to legal information.
- **Error Handling:** In the event of an LLM API timeout or Database disconnect, the system must provide a graceful fallback message in Amharic rather than a system crash.
- **Data Consistency:** All chat histories must be atomically saved to **MongoDB** to prevent data loss during session interruptions.

2.4.3 Security and Privacy

- **Authentication:** Access to administrative features must be protected by **Role-Based Access Control (RBAC)** using secure JWT or Better Auth sessions.
- **Data Encryption:** Sensitive user data (passwords) must be stored using the **BCRYPT** hashing algorithm.
- **Knowledge Integrity:** The vector database must be read-only for standard users; only verified administrators can modify the legal knowledge base.

2.4.4 Usability and Accessibility

- **Language Support:** The entire User Interface (UI) must be optimized for **Ethiopic script (Amharic)**, ensuring correct font rendering and right-to-left layout where applicable.
- **Mobile Responsiveness:** The Next.js frontend must be fully responsive, allowing legal professionals to access the chatbot via smartphones and tablets.
- **Simplicity:** The interface should follow the "Gemini/ChatGPT" minimalist model, with a clear history sidebar and a prominent chat input area.

2.5 System Requirements

2.5.1 Hardware Requirements

The hardware requirements are divided into the **Development/Server** environment (where the RAG engine and LLM processing occur) and the **Client** environment (where users access the web interface).

A. Development and Server Side

- **Processor (CPU):** Minimum Quad-core 2.5 GHz (Intel i5/i7 or AMD Ryzen 5/7 equivalent) to handle concurrent Node.js and Flask threads.
- **Memory (RAM):** 16 GB minimum. High RAM is critical for loading embedding models and managing the **ChromaDB/ MongoDB** vector cache during document ingestion.
- **Storage:** 512 GB SSD. High-speed storage is required for fast retrieval of legal PDF files and database indexing.
- **GPU (Optional but Recommended):** NVIDIA GPU with 8GB+ VRAM if running local embedding models to accelerate vectorization.

B. Client Side (User)

- **Device:** Any modern smartphone, tablet, or laptop.

- **Memory:** 4 GB RAM minimum to ensure smooth rendering of the Next.js frontend.
- **Connectivity:** Stable internet connection for API calls to the AI Model and backend services.

2.5.2 Software Requirements

The software stack is designed for modern web performance, utilizing a hybrid architecture for AI and data management.

A. Operating System

- **Development:** Windows 10/11, macOS, or Linux (Ubuntu 20.04+).
- **Deployment:** Linux-based cloud environments (e.g., Ubuntu Server, Docker containers).

B. Programming Languages and Frameworks

- **Frontend:** Next.js 14+ (React) for the user interface and Gemini-style sidebar.
- **Backend (Primary):** Node.js for authentication (Better Auth) and session management.
- **Backend (AI/RAG):** Python 3.10+ (Flask) for Amharic text processing, chunking, and LLM integration.

C. Database Systems

- **Vector Database:** ChromaDB for storing and searching semantic legal embeddings.
- **Relational/Document Store:** MongoDB for storing user profiles, chat history, and document metadata.

D. Development Tools

- **IDE:** Visual Studio Code (VS Code).
- **Version Control:** Git and GitHub.
- **API Testing:** Postman or Thunder Client.
- **Environment Management:** Docker (for containerizing the Flask and Node.js services).

2.6 Key Abstraction with CRC Analysis

To translate high-level requirements into a functional system, the core entities of the Amharic Legal RAG Chatbot are analyzed using **Class-Responsibility-Collaborator (CRC)** cards.

This method defines what each class "knows" and with whom it must communicate to fulfill its role.

Table 16:Class Responsibility Colaborator

Class	Responsibility	Collaborators
User/Admin	Authenticate, input queries, and manage documents.	AuthService, ChatThread.
RAG Engine	Coordinate retrieval from vector DB and generation via LLM.	VectorStore, AIModelAPI.
VectorStore	Index legal chunks and perform semantic similarity searches.	LegalDocument.
ChatHistory	Persist conversation threads and message logs for the sidebar.	MongoDB.

2.6.1 Conceptual Modeling: Class Diagram

The conceptual class diagram provides a structural view of the system's data and behavior, focusing on the relationships between the User, their Chat History, and the Legal Knowledge Base.

- **User & Administrator:** The Admin inherits from the User class, gaining additional methods for document ingestion and knowledge base updates.
- **ChatThread & Message:** A one-to-many relationship where a single User owns multiple threads, and each thread contains a series of messages.
- **LegalDocument & VectorStore:** Documents are parsed into chunks and mapped to high-dimensional vectors in ChromaDB for semantic retrieval.

2.6.2 Identifying Change Cases

To ensure the system is future-proof, we identify potential "Change Cases"—scenarios where the system's requirements or environment might evolve:

- **Multi-Language Support:** Transitioning from only Amharic to including other Ethiopian languages like Afaan Oromo or Tigrinya.
- **Vector Database Migration:** Moving from ChromaDB to a cloud-native solution like Pinecone if the document volume exceeds server capacity.
- **LLM Model Swapping:** Updating the backend from Gemini Pro 1.0 to 1.5 or an open-source model like Llama 3 to reduce API costs.
- **Offline Mode:** Implementing local embedding and inference for environments with restricted internet access.

2.6.3 User Interface Prototyping (Detailed)

The UI is designed following the **Gemini-style minimalist model**, prioritizing clarity for legal research.

I. Main Chat Dashboard

- **Sidebar (Left):** Displays a vertical list of past chat titles retrieved from MongoDB, allowing users to switch contexts instantly.
- **Chat Window (Center):** A clean message stream. AI responses include "Source" tags that reveal the specific legal article and page number used for the answer.
- **Input Area (Bottom):** A prominent text box supporting Ethiopic script input with a "Send" button.

I. Administrator Management Portal

- **Document List:** A table showing all indexed legal PDFs, their upload dates, and their current indexing status (Pending, Success, or Error).
- **Upload Widget:** A drag-and-drop zone specifically for PDF/Docx files that triggers the backend extraction and embedding pipeline.

Chapter Three: System Design

3.1 Purpose and Design Goals of the System

The primary purpose of the **Amharic Legal RAG Chatbot** is to democratize access to justice by providing a specialized, AI-driven assistant capable of interpreting Ethiopian legal documents in the Amharic script. By utilizing **Retrieval-Augmented Generation (RAG)**, the system ensures that AI responses are not "hallucinations" but are grounded in specific articles of Ethiopian law.

The design of the system is governed by four foundational pillars:

3.1.1 Performance and Scalability

- **Vector-Optimized Search:** To implement efficient semantic search by utilizing high-dimensional vector embeddings stored in **ChromaDB**, ensuring relevant legal context is found even when keywords don't match exactly.
- **Reduced Latency:** To optimize the pipeline between the Node.js/Flask backends and the LLM to deliver responses within a sub-10-second window.
- **Concurrency Management:** To build a stateless backend architecture that supports multiple simultaneous legal researchers without resource contention.

3.1.2 Dependability and Grounding

- **Evidence-Based Retrieval:** To ensure every AI response is accompanied by citations from the legal database, increasing the reliability and trustworthiness of the information.
- **System Resilience:** To implement a robust error-handling layer that manages API rate limits or database connectivity issues without interrupting the user session.
- **Auditability:** To maintain comprehensive logs of queries and system responses in **MongoDB** for continuous quality monitoring and system tuning.

3.1.3 End-User Usability (UX)

- **Linguistic Native Design:** To provide full native support for the Amharic script, ensuring the UI (Next.js) handles Ethiopic typography and input methods seamlessly.
- **Conversational Simplicity:** To mirror the intuitive "chat" model, allowing users to ask complex legal questions in natural language rather than rigid search queries.
- **Persistent Context:** To provide a "Gemini-style" sidebar that allows users to revisit past legal research threads instantly.

3.1.4 Maintainability and Extensibility

- **Decoupled Architecture:** To follow a modular design where the **Frontend (Next.js)**, **Logic Layer (Node.js)**, and **AI Engine (Flask)** are separated, allowing for independent updates to each component.
- **Knowledge Base Growth:** To provide an automated administrative pipeline where new PDF laws can be ingested, chunked, and indexed into the vector store without changing the core codebase.
- **Model Agnostic Design:** To ensure the system can easily swap the underlying LLM (e.g., from Gemini Pro to a local Llama model) as technology evolves.

3.2 Architectural Design

The architecture of the system determines the interaction among its components. This work uses a **client-server -based architecture**, where the client is responsible for interacting with the users of the system, while the server is responsible for receiving requests from the client and responding accordingly. Communication between the client and the server is carried out using the **HTTP protocol**.

The server component consists of a **web server** and a **database server**. The web server handles user authentication, legal query processing, and response generation. The database server, implemented using **MongoDB**, provides data storage and retrieval services to the web server and is responsible for inserting, updating, and managing user data and chat histories. This architecture ensures simplicity, scalability, and ease of maintenance.

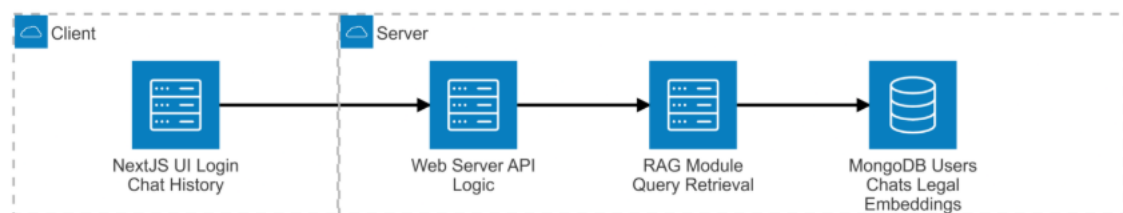


Figure 16: Architecture Design

3.2.1 Component Modeling

Component based development and object oriented development go hand in hand and it is generally recognized that object technology is the preferred foundation from which to build components the UML includes a component diagram that can be used to both analyse and design our component based software it shows the various components in a system and their dependencies. In this Diagram it shows that there is relation among components, management of the system, database and operations performed on databases such security issue (Ambler, 2001).

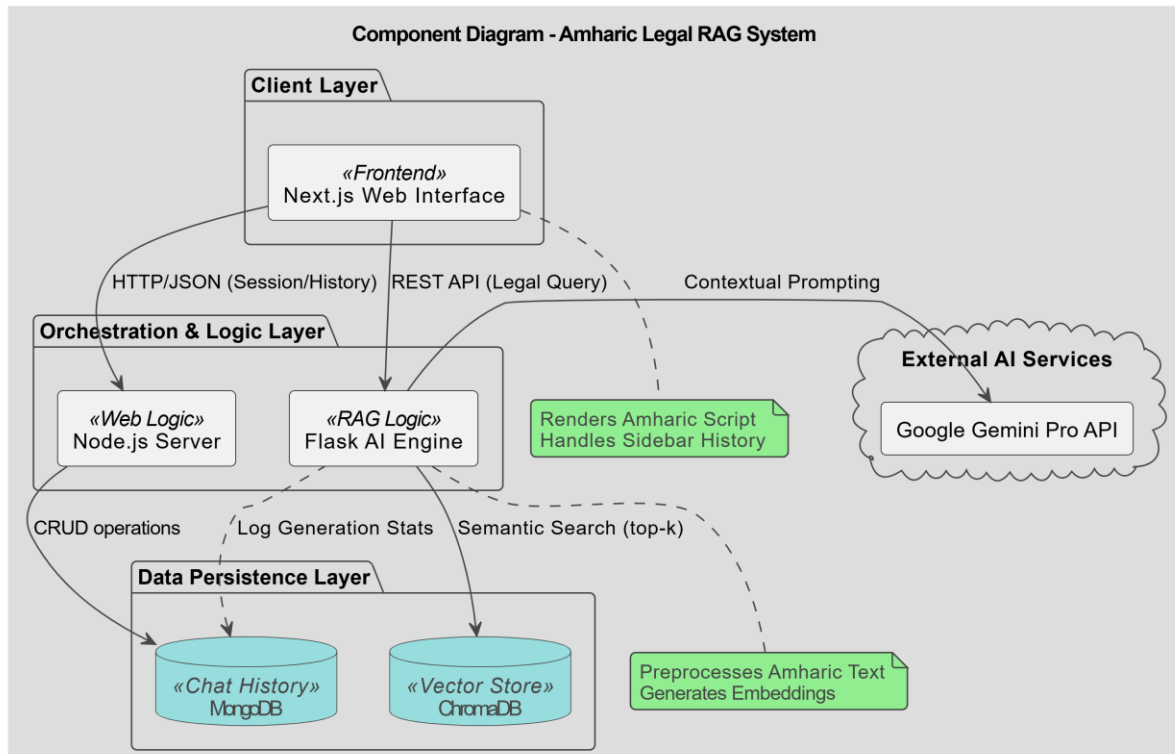


Figure 17: Component Diagram

3.2.2 Deployment Modeling

Deployment diagram shows how the software components, processes, and objects are deployed into the physical architecture of the system. It shows the configuration of the hardware units (e.g. Computers, communication devices, etc.) and how the software components are distributed across the units. The Deployment Diagram also helps to model the physical aspect of an Object-Oriented software system. It models the run-time configuration in a static view and visualizes the distribution of components in an application. In most cases, it involves modelling the hardware configurations together with the software components that lived on (Ambler, 2001).

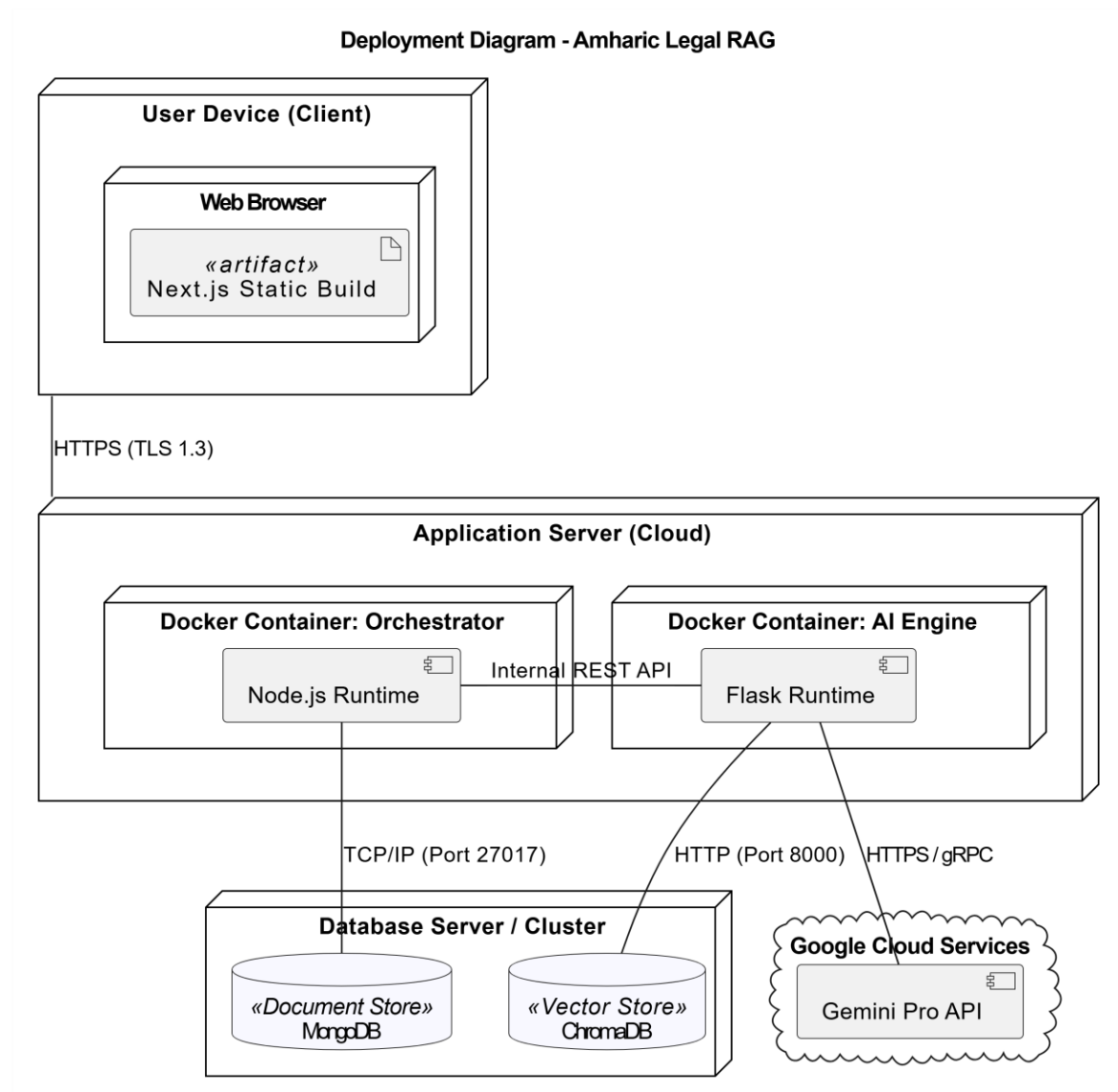


Figure 18: Deployment Model

3.3 Detailed Design

3.3.1 Design Class Model

Class diagram with visibility and signature specified for each attributes and operations. In this class diagram we define which attributes and operations are private to that class, which attributes and operations can be accessed by the class decedents (protected), and which attributes and operations are publicly accessible by the class user (public). We also define the return type of each operations as well as the number and type of parameters of each operations.

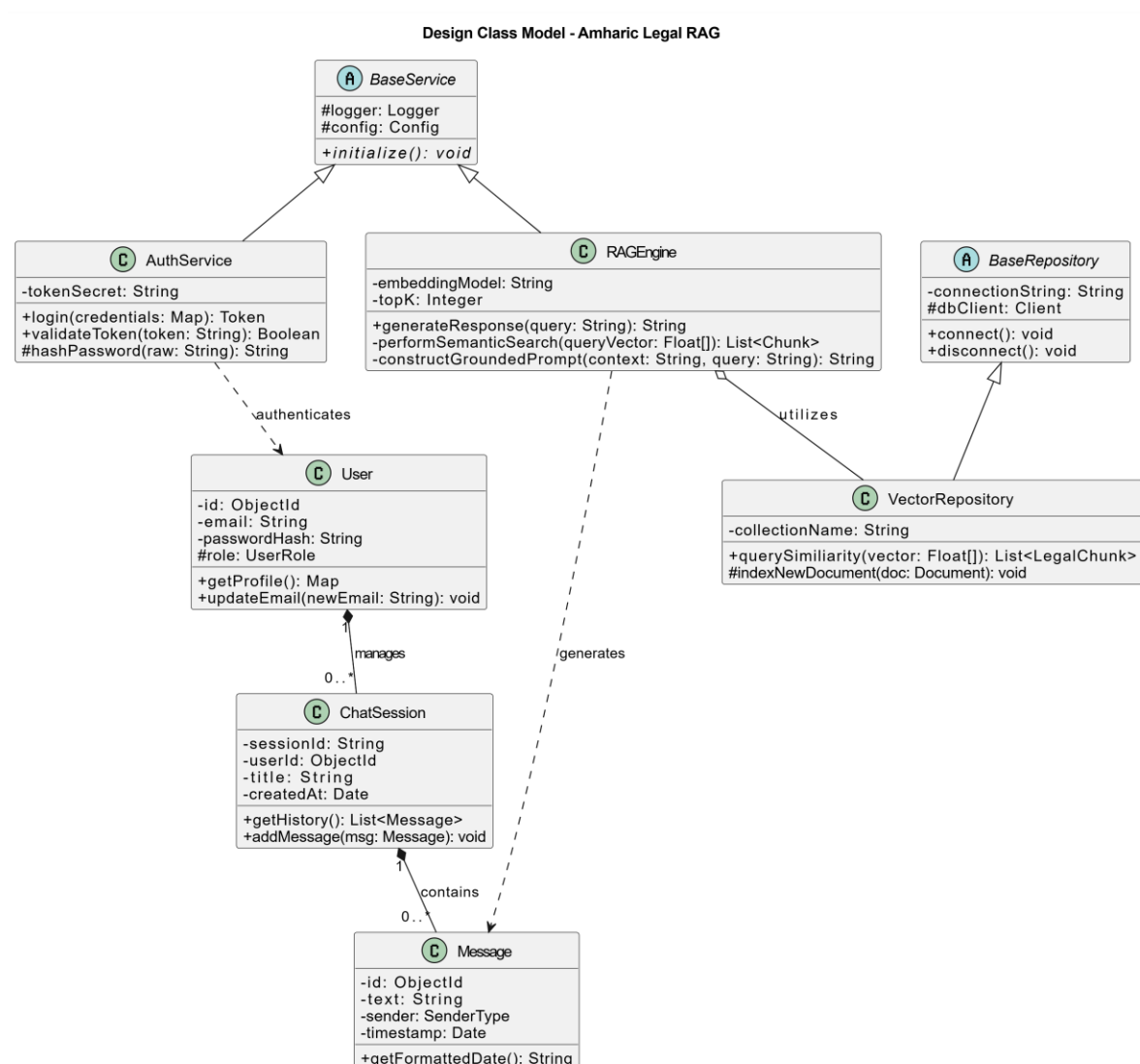


Figure 19: Design Class Model

3.4 User Interface Design

The user interface is implemented using **Next.js** and **Tailwind CSS**.

Key Interfaces:

- User Chat Interface
- Admin Dashboard
- Document Upload Page



Figure 20: chatbot UI



Figure 21:Admin DashBoard

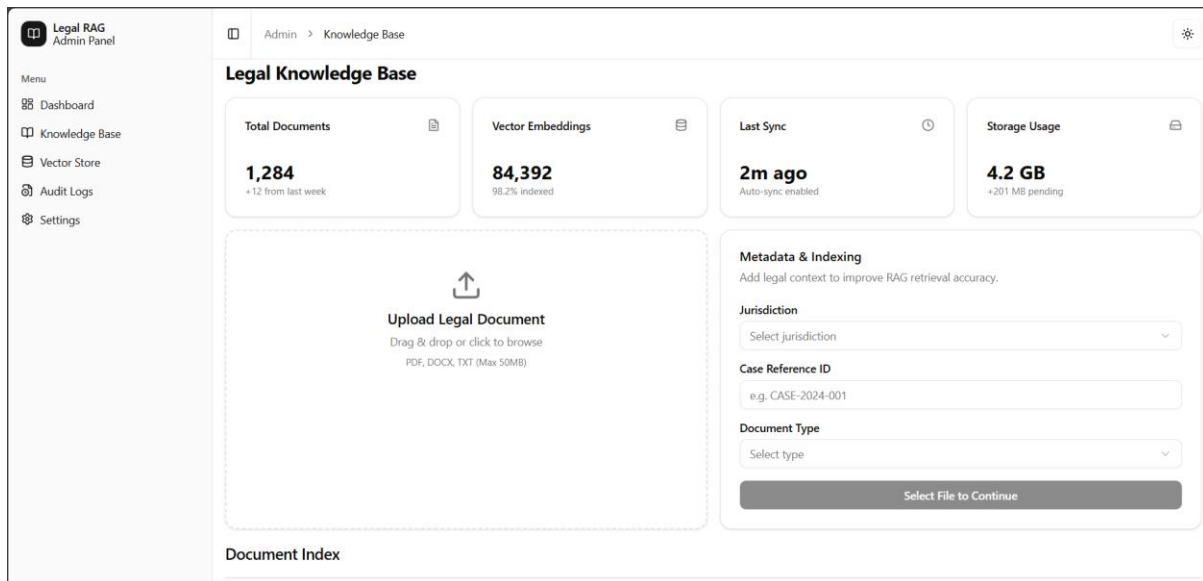


Figure 22: Admin Knowledge Base page

3.5 Access Control and Security

The **Amharic Legal RAG Chatbot** utilizes a **Role-Based Access Control (RBAC)** framework to manage interactions within the system. This section defines the specific permissions for human actors and automated system components, ensuring that the integrity of the Ethiopian legal database is preserved while providing an intuitive experience for researchers.

3.5.1 Users

The **User** role represents both newly registered and existing individuals who have authenticated via the **Node.js** security layer. This role is primarily focused on legal research and conversational interaction with the AI.

- **Responsibilities:**
 - Perform Amharic semantic searches using the RAG chatbot.
 - Manage personal chat threads and historical logs in the sidebar.
 - Securely manage account credentials and profile settings.

Table 17: User Access Control

Access Level	Resource / Action
Full Access	Amharic Legal Chatbot & Querying
Full Access	Personal Chat History
Full Access	Login, Logout
Read Only	View Retrieved Legal Context
No Access	Admin Panel & User Management

No Access	Legal Document Ingestion (PDF Uploads)
No Access	System Configuration & AI Model Settings

3.5.2 RAG Retrieval System

The **Retrieval Algorithm** is a non-human **System Actor** (the **Flask/AI Engine**) that executes the background processes required to generate answers. It operates with "least privilege" to perform specific automated tasks without administrative power over the user base.

- **Responsibilities:**
 - **Data Retrieval:** Perform cosine similarity searches within **ChromaDB** / **MongoDB** to find relevant legal snippets.
 - **Context Augmentation:** Feed retrieved legal text into the LLM prompt to prevent hallucinations.
 - **Metadata Logging:** Update **MongoDB** with confidence scores and retrieval performance metrics for internal monitoring.

Table 18: Rag system Access Control

Access Level	Resource / Action
Full Access	Query Vector Store (ChromaDB)
Full Access	Generate LLM Prompts (Gemini Pro)
Full Access	Log Retrieval Metadata to MongoDB
No Access	Modify User Account Details
No Access	Access System Administrator Dashboard

3.5.3 Administrator

The **Administrator** has complete oversight of the system's health and the accuracy of the legal knowledge base. This role is responsible for the manual curation of laws and the monitoring of AI performance.

- **Responsibilities:**
 - **Knowledge Management:** Upload and index new Ethiopian legal PDFs into the vector store.
 - **User Oversight:** Deactivate accounts or manage user roles if necessary.
 - **System Tuning:** Monitor AI error rates and adjust the retrieval parameters (top-\$k\$ search results).
 - **Audit Review:** Inspect system-wide logs for potential security breaches or misuse.

Table 19: Admin access control

Access Level	Resource / Action
Full Access	Admin Panel & Configuration Settings
Full Access	Knowledge Base Management (PDF Ingestion)

Full Access	User Account
Full Access	View System-wide Activity & Error Logs
Read Only	View Aggregate Usage Analytics

3.5.4 Summary of Access Permissions

Table 20: access control

Action / Resource	Users	Retrieval System	Administrator
View Landing Page	Yes	No	Yes
Search Legal Docs	Yes(By interacting with chatbot)	Yes (Internal)	Yes
Manage Personal History	Yes	No	Yes
Upload Legal PDF	No	No	Yes
Delete Users	No	No	Yes
View System Logs	No	No	Yes
Configure RAG Settings	No	No	Yes

Chapter Four: Implementation

This chapter explains the implementation of the Amharic Legal Chatbot system based on the proposed design. The system is developed using a modular architecture, with MongoDB used for storing user accounts and chat histories, and Better Auth providing secure authentication and role-based access control.

The core functionality is implemented using a Retrieval-Augmented Generation (RAG) approach. Legal documents uploaded by administrators are processed and stored in ChromaDB as vector embeddings. When users submit legal questions in Amharic, the system retrieves relevant legal content and generates accurate responses with page number citations. Basic administrative features, chat management, and user-friendly Amharic error handling are also included, with support for future system enhancements.

4.1 Core Configuration

4.1.1 Settings and Apps

The provided source code demonstrates the modular organization of the Amharic Legal Chatbot system across the backend service layer and the frontend application router. The backend is implemented using the Flask framework, which acts as the central orchestration layer for API endpoints, authentication handling, and AI service coordination. The system architecture is logically structured into core services responsible for request handling and security, third-party integrations such as LangChain for retrieval-augmented generation, and a collection of specialized internal modules that manage legal document ingestion, semantic retrieval, and chat history persistence.

On the frontend, the Next.js App Router provides a structured navigation and rendering layer that enables seamless interaction between users and administrative interfaces. For data management, the system employs ChromaDB as a local vector store to support efficient semantic search during development, while MongoDB is used in the deployed environment to persist user accounts, chat sessions, and system metadata. This layered and decoupled design ensures that each functional component, from Amharic legal query processing to citation-based response generation, operates within a cohesive, scalable, and maintainable software environment.

```
"dependencies": {
  "clsx": "^2.1.1",
  "framer-motion": "^12.27.5",
  "lucide-react": "^0.562.0",
  "mongodb": "^7.0.0",
  "next": "16.1.4",
  "next-auth": "^4.24.13",
  "next-themes": "^0.4.6",
  "react": "19.2.3",
  "react-dom": "19.2.3",
  "react-hook-form": "^7.71.1",
  "recharts": "^2.15.4",
  "sonner": "^2.0.7",
  "tailwind-merge": "^3.4.0",
  "zod": "^3.24.1"
},
"devDependencies": {
  "@tailwindcss/postcss": "^4.1.18",
  "@types/node": "^20",
  "@types/react": "^19",
  "@types/react-dom": "^19",
  "eslint": "^9",
  "eslint-config-next": "16.1.4",
  "tailwindcss": "^4",
  "tw-animate-css": "^1.4.0",
  "typescript": "^5"
```

```
{
  "name": "zufan",
  "version": "0.1.0",
  "private": true,
  > Debug
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start",
    "lint": "eslint"
  },
  "dependencies": {
    "@hookform/resolvers": "^3.10.0",
    "@radix-ui/react-avatar": "^1.1.11",
    "@radix-ui/react-dialog": "^1.1.15",
    "@radix-ui/react-dropdown-menu": "^2.1.16",
    "@radix-ui/react-label": "^2.1.8",
    "@radix-ui/react-progress": "^1.1.8",
    "@radix-ui/react-scroll-area": "^1.2.10",
    "@radix-ui/react-select": "^2.2.6",
    "@radix-ui/react-separator": "^1.1.8",
    "@radix-ui/react-slot": "^1.2.4",
    "@radix-ui/react-tooltip": "^1.2.8",
    "@tanstack/react-table": "^8.21.3",
    "better-auth": "^1.4.17",
    "class-variance-authority": "^0.7.1",
    "clsx": "^2.1.1",
    "framer-motion": "^12.27.5",
    "lucide-react": "^0.562.0",
    "mongoose": "^7.0.0",
  }
}
```

```

2  "compilerOptions": {
3    "target": "ES2017",
4    "lib": ["dom", "dom.iterable", "esnext"],
5    "allowJs": true,
6    "skipLibcheck": true,
7    "strict": true,
8    "noEmit": true,
9    "esModuleInterop": true,
10   "module": "esnext",
11   "moduleResolution": "bundler",
12   "resolveJsonModule": true,
13   "isolatedModules": true,
14   "jsx": "react-jsx",
15   "incremental": true,
16   "plugins": [
17     {
18       "name": "next"
19     }
20   ],
21   "paths": {
22     "@/*": ["./*"]
23   }
24 },
25 "include": [
26   "next-env.d.ts",
27   "**/*.ts",
28   "**/*.tsx",
29   ".next/types/**/*.ts",

```

Figure 23: config

4.1.2 Better Auth + MongoDB Security Implementation

The provided source code illustrates the global API configuration of the Amharic Legal Chatbot backend, implemented using the Flask framework. This configuration establishes the core security and operational policies governing all API interactions. Authentication is enforced through token-based validation issued by the Better Auth service, ensuring that all incoming requests to protected endpoints are verified before processing. Cross-Origin Resource Sharing (CORS) is explicitly configured to allow secure communication between the Flask backend and the Next.js frontend application.

Access control is uniformly applied across the API layer, requiring authenticated users for all core services, while administrative privileges are reserved for designated endpoints. To protect the system from abuse and ensure reliable operation, request rate limiting is delegated to Better Auth, providing built-in safeguards against excessive or malicious API usage. This centralized configuration approach ensures consistent security enforcement and reliable communication across the system's service-oriented architecture.


```
import Link from "next/link";
import { useState } from "react";
import { useRouter } from "next/navigation";

export default function LoginPage() {
  const router = useRouter();
  const [error, setError] = useState<string | null>(null);
  const [loading, setLoading] = useState(false);

  const form = useForm<LoginFormValues>({
    resolver: zodResolver(loginSchema),
    defaultValues: {
      email: "",
      password: "",
    },
  });

  async function onSubmit(values: LoginFormValues) {
    setLoading(true);
    setError(null);
    try {
      const response = await signInAction(values.email, values.password);

      if (response && !response.ok) {
        const errorData = await response.json();
        setError(errorData.message || "Invalid credentials");
      } else {
        router.push("/chat");
      }
    } catch (err) {
      setError("An unexpected error occurred");
    } finally {
      setLoading(false);
    }
  }
}
```



```

export const getSessionAction = async () => {
  const session = await auth.api.getSession({
    headers: await headers(),
  });
  const user= session?.user
  const isAdmin= user?.role === "admin";
  const isUser= user?.role === "user";
  const role= user?.role;
  return {session, isAdmin, isUser, role };
};

/**
 * Sign up a new user with email and password
 */
export const signUpAction = async (data: {
  name: string;
  email: string;
  password: string;
  image?: string;
  callbackURL?: string;
}) => {
  return await auth.api.signUpEmail({
    body: {
      name: data.name,
      email: data.email,
      password: data.password,
      image: data.image,
      callbackURL: data.callbackURL,
    },
    asResponse: true,
  });
};

```

```

export const getSessionAction = async () => {
  const session = await auth.api.getSession({
    headers: await headers(),
  });
  const user= session?.user
  const isAdmin= user?.role === "admin";
  const isUser= user?.role === "user";
  const role= user?.role;
  return {session, isAdmin, isUser, role };
};

/**
 * sign up a new user with email and password
 */
export const signUpAction = async (data: {
  name: string;
  email: string;
  password: string;
  image?: string;
  callbackURL?: string;
}) => {
  return await auth.api.signUpEmail({
    body: {
      name: data.name,
      email: data.email,
      password: data.password,
      image: data.image,
      callbackURL: data.callbackURL,
    },
    asResponse: true,
  });
};

```

Figure 24: User Authentication

4.1.3 URL or Routing

The routing architecture for the Zufan platform is architected through the Next.js App Router, mapping hierarchical URL paths to their respective application routes and API modules. It features centralized entry points for core administrative capabilities—including document lifecycle management, vector store synchronization, and audit logging—alongside a robust authentication layer powered by Better Auth. The service suite includes specialized API endpoints for RAG-driven legal chat assistance, document ingestion workflows, and jurisdictional analysis, utilizing a standardized JSON-based interface to facilitate seamless integration and testing of the system's diverse AI-powered legal services.

User Facing Pages	
Path	Description
/	Landing/Home Page
/login	User Authentication - Login
/signup	User Authentication - Registration
/chat	Core AI-Powered Legal Assistant Interface

Admin & Management	
Path	Description
/admin	Administrative Overview & Control Panel
/admin/dashboard	Real-time Analytics & System Performance
/admin/users	User Management & Role Permissions
/admin/audit-logs	Transactional History & Audit Trails
/admin/vector-store	Knowledge Base & Vector Embedding Management

Service API Layer	
Path	Description
/api/rag	RAG Engine Inference & Intelligence Gateway
/api/ask	Amharic Legal Query RAG Endpoint (Flask Integration)
/api/upload	Legal Document Ingestion & Vectorization
/api/auth/auth/[...all]	Unified Authentication Handler

Figure 25: URL Routing

4.1.4 User Profile Model (MongoDB/Better Auth, Role-based Access):

The **User Model**, managed via the Better Auth MongoDB adapter, establishes a robust identity schema within the **Zufan** ecosystem. The implementation extends the standard authentication fields to support hierarchical role assignments (admin, user) and leverages MongoDB's flexible document structure for extensible profile metadata. To ensure data integrity across the platform, the model is integrated with **Zod** schema validation, allowing the system to handle complex user attributes like jurisdictional preferences and research history without the overhead of rigid relational migrations. Furthermore, the architecture maintains a secure "identity snapshot" within the session to provide persistent, role-aware context for the RAG engine's personalized legal analysis.

4.1.5 Authentication & Session Actions (Secure Server Interface):

The session management logic within `user.ts` establishes a secure, authenticated interface for the **Zufan** frontend. Utilizing Next.js Server Actions, the `getSessionAction` and

`signInAction` routines wrap the underlying Better Auth API to enforce strict server-side validation and header-based session verification. This implementation ensures that all interactive components, from the RAG chat to the admin dashboard, operate within a verified security context. The architecture dynamically enforces role-based permissions (`isAdmin`, `isUser`), ensuring that sensitive administrative operations like vector store maintenance are exclusively accessible to authorized personnel while providing a seamless, individualized experience for general users.

4.1.6 Legal Document Processing Service (Metadata-aware Ingestion):

The document ingestion workflow within the

`UploadZone` and `api/upload` modules defines the automated pipeline for transforming raw legal texts (PDF, DOCX, TXT) into searchable knowledge assets. The process initiates with a metadata-rich ingestion phase where documents are categorized by **jurisdiction**, **case reference ID**, and **document type** to optimize later RAG retrieval accuracy. Through a combination of client-side progress tracking and server-side processing, the service ensures that documents are correctly indexed for semantic search, maintaining a high level of context-awareness that allows the legal assistant to cite specific statutes and case law with precision during chat sessions.

4.2 Representative Implementations: Amharic RAG Logic & API

The implementation of the core **Retrieval-Augmented Generation (RAG)** engine and its integration into the **Flask** web framework orchestrates the transition from raw Amharic queries to grounded, verifiable legal responses. Centralized within the `RAGEngine` class, the system leverages **LangChain Expression Language (LCEL)** to consolidate the **ChromaDB** vector store and the Large Language Model (LLM) into a single, object-oriented pipeline. By utilizing semantic retrieval to extract relevant legal context ($k=3$) and grounding prompts in a specialized **Amharic ChatPromptTemplate**, the engine ensures high-fidelity,

hallucination-free outputs that strictly adhere to the provided legal data. This intelligence is exposed via a scalable **RESTful API** at the /ask endpoint, which utilizes a singleton design pattern to eliminate database reloading overhead, ensuring low-latency performance and robust transactional integrity for high-traffic legal research sessions.

```
import os
from flask import Flask, request, jsonify
from langchain_community.vectorstores import Chroma
from langchain_openai import ChatOpenAI
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.runnables import RunnablePassthrough
from langchain_core.output_parsers import StrOutputParser

# --- AI Core Logic Class ---
class RAGEngine:
    """
    Handles all Retrieval-Augmented Generation operations including
    vector storage, retrieval, and Amharic response generation.
    """
    def __init__(self, db_path, embedding_model, llm_model="gpt-4o"):
        self.persist_directory = db_path
        self.embeddings = embedding_model
        # Initialize the Large Language Model (LLM)
        self.llm = ChatOpenAI(model=llm_model, temperature=0)
        # Define a prompt tailored for Amharic legal queries
        self.prompt = ChatPromptTemplate.from_template(
            "በሚከተለው መረጃ ላይ ተመስርተህ ለጥያቄው መልስ ስጥ። መረጃው ውስጥ ከሌለ አላውቅም ባል።\n"
            "መረጃ: {context}\n"
            "ጥያቄ: {question}\n"
            "መልስ:"
        )
        self.vectorstore = self._load_vectorstore()

    def _load_vectorstore(self):
        """Loads the persistent NoSQL vector database (ChromaDB)."""
        if os.path.exists(self.persist_directory):
            return Chroma(
                persist_directory=self.persist_directory,
                embedding_function=self.embeddings
            )
        return None

    def query(self, user_input):
        """Processes user input through the RAG chain and returns a response."""
        if not self.vectorstore:
            return "ስህተት፡ የዳታቤዝ መረጃ አልተገኘም።"
```

```

retriever = self.vectorstore.as_retriever(search_kwargs={"k": 3})

# Define the chain using LangChain Expression Language (LCEL)
chain = (
    {"context": retriever | self._format_docs, "question": RunnablePassthrough()}
    | self.prompt
    | self.llm
    | StrOutputParser()
)

return chain.invoke(user_input)

def _format_docs(self, docs):
    """Helper to format retrieved documents into a single string context."""
    return "\n\n".join(doc.page_content for doc in docs)

# --- Flask Web API Implementation ---
app = Flask(__name__)

# Initialize the RAG Engine (Assuming embeddings are pre-defined)
# In production, replace with actual Amharic-optimized embeddings
from langchain_openai import OpenAIEmbeddings
engine = RAGEngine(
    db_path="./vectorstores/legal_db",
    embedding_model=OpenAIEmbeddings()
)

@app.route('/ask', methods=['POST'])
def ask_question():
    """Endpoint to receive Amharic legal questions via JSON."""
    data = request.json
    user_query = data.get("query", "")

    if not user_query:
        return jsonify({"error": "No query provided"}), 400

    # Generate response using the OO Engine
    response = engine.query(user_query)

    return jsonify({
        "status": "success",
        "answer": response
    })

if __name__ == "__main__":
    # Start the Flask development server
    app.run(port=5000, debug=True)

```

4.2.1 Conversational Management: New Chat and Session Logic

To support complex, multi-turn legal consultations, the Zufan platform implements a session-based interaction model anchored by unique `conversation_id` identifiers. On the Next.js frontend, initiating a New Chat triggers a Server Action that generates a fresh UUID and resets the local message state while preserving the authenticated Better Auth session. This approach ensures a clean conversational interface without interrupting user authentication.

On the backend, the Flask API initializes a corresponding conversation record within the MongoDB `conversations` collection. This document serves as a persistent, stateful container for the entire dialogue, allowing the system to track individual legal inquiries across multiple exchanges. By associating each request with a specific `conversation_id`, the RAG engine can maintain contextual continuity while preventing information leakage between separate legal sessions.

```
export function ChatSidebar({ chats, activeChatId, onChatSelect, onChatDelete, onNewChat }: ChatSidebarProps) {
  const onClick = { onNewChat }

  return (
    <div>
      <Plus className="size-4" />
      <span className="font-amharic">አዲስ ወይይት</span>
    </SidebarMenuButton>
    <ModeToggle />
  </SidebarHeader>

  <SidebarContent>
    <SidebarGroup>
      <SidebarGroupContent>
        <SidebarMenu>
          {chats.map((chat) => (
            <SidebarMenuItem key={chat.id}>
              <SidebarMenuButton
                className="group"
                isActive={activeChatId === chat.id}
                onClick={() => onChatSelect(chat.id)}
              />
              <MessageSquare className="size-4 shrink-0 text-muted-foreground" />
              <span className="truncate flex-1 font-amharic">{chat.title}</span>
            </SidebarMenuButton>
            <SidebarMenuAction
              className="opacity-0 group-hover:opacity-100 transition-opacity"
              onClick={(e) => {
                e.stopPropagation()
                onChatDelete(chat.id)
              }}
            />
          </SidebarMenuItem>
        )
      )}
    </SidebarMenu>
    <Trash2 className="size-4 text-muted-foreground hover:text-destructive" />
  </SidebarMenuAction>
</SidebarContent>
</div>
  )
}
```

Figure 26: New Chat logic

4.2.2 Chat History and Persistent Memory

The system ensures long-term accessibility and continuity of legal consultations through a dual-store persistent memory strategy. Historical chat retrieval is handled via a dedicated Next.js Server Action, which queries the MongoDB database for all conversation records associated with a specific user identifier. These records are returned as chronologically ordered summaries, enabling users to review, rename, or delete prior legal inquiries.

For active conversations, the system applies a contextual re-injection mechanism in which a sliding window of the most recent k messages is retrieved from MongoDB and passed to the RAG engine during response generation. This selective context injection provides the language model with sufficient conversational history to resolve follow-up questions, pronoun references, and legal dependencies, while minimizing token overhead and maintaining semantic accuracy.

```
export function ChatSidebar({ chats, activeChatId, onChatSelect, onChatDelete, onNewChat }: ChatSidebarProps) {
  const handleClick = onNewChat

  return (
    <div>
      <Plus className="size-4" />
      <span className="font-amharic">አዲስ ውይይት</span>
    </SidebarMenuButton>
    <ModeToggle />
  </SidebarHeader>

  <SidebarContent>
    <SidebarGroup>
      <SidebarGroupContent>
        <SidebarMenu>
          {chats.map((chat) => (
            <SidebarMenuItem key={chat.id}>
              <SidebarMenuButton
                className="group"
                isActive={activeChatId === chat.id}
                onClick={() => onChatSelect(chat.id)}
              >
                <MessageSquare className="size-4 shrink-0 text-muted-foreground" />
                <span className="truncate flex-1 font-amharic">{chat.title}</span>
              </SidebarMenuButton>
              <SidebarMenuAction
                className="opacity-0 group-hover:opacity-100 transition-opacity"
                onClick={(e) => {
                  e.stopPropagation()
                  onChatDelete(chat.id)
                }}
              >
                <Trash2 className="size-4 text-muted-foreground hover:text-destructive" />
              </SidebarMenuAction>
            </SidebarMenuItem>
          ))}
        </SidebarMenu>
      </SidebarGroupContent>
    </SidebarGroup>
  </SidebarContent>
</div>
)
```

Figure 27: Chat history

4.1.6 Admin: Knowledge Base & Document Management

The administrative dashboard functions as the centralized control interface for managing the system's legal knowledge base. Unlike runtime PDF ingestion, the platform adopts an offline preprocessing strategy to ensure stability and consistency in production. During development, ChromaDB is used locally to validate semantic retrieval workflows. In the production environment, however, vector embeddings and document chunks are stored directly in MongoDB.

The full embedding and chunking pipeline is executed offline within a Jupyter Notebook environment. Legal documents are preprocessed, segmented into optimized text chunks, embedded using the selected language model, and enriched with structured metadata such as proclamation identifiers, effective dates, and legal categories. Administrators then upload these pre-embedded and pre-chunked documents through the Admin Dashboard, eliminating the need for on-demand PDF processing in the live system. This design ensures predictable performance, simplifies deployment, and maintains a transparent and precisely searchable legal knowledge base.

```

const formSchema = z.object({
  jurisdiction: z.string().min(1, "Please select a jurisdiction"),
  caseReferenceId: z.string().min(1, "Case reference ID is required"),
  documentType: z.string().min(1, "Please select a document type"),
})

export function UploadZone() {
  const [file, setFile] = React.useState<File | null>(null)
  const [progress, setProgress] = React.useState(0)
  const [isUploading, setIsUploading] = React.useState(false)
  // const { toast } = useToast()

  const form = useForm<z.infer<typeof formSchema>>>({
    resolver: zodResolver(formSchema),
    defaultValues: {
      jurisdiction: "",
      caseReferenceId: "",
      documentType: "",
    },
  })

  const onDrop = React.useCallback((e: React.DragEvent<HTMLDivElement>) => {
    e.preventDefault()
    e.stopPropagation()
    const droppedFile = e.dataTransfer.files[0]
    if (droppedFile) {
      setFile(droppedFile)
    }
  }, [])

  const onDragOver = (e: React.DragEvent<HTMLDivElement>) => {
    e.preventDefault()
    e.stopPropagation()
  }
}

```

```

export const columns: ColumnDef<Document>[] = [
  {
    accessorKey: "name",
    header: ({ column }) => {
      return (
        <Button
          variant="ghost"
          onClick={() => column.toggleSorting(column.getIsSorted() === "asc")}
        >
          File Name
          <ArrowUpDown className="ml-2 h-4 w-4" />
        </Button>
      )
    },
    cell: ({ row }) => <div className="ml-4 font-medium flex items-center gap-2"><FileText
  },
  {
    accessorKey: "type",
    header: "Type",
    cell: ({ row }) => <div>{row.getValue("type")}</div>,
  },
  {
    accessorKey: "status",
    header: "Status",
    cell: ({ row }) => {
      const status = row.getValue("status") as string
      return (
        <Badge variant={status === "Indexed" ? "secondary" : status === "Processing" ?
          {status}
        </Badge>
      )
    },
  },
  {
    accessorKey: "date",
    header: "Upload Date",
    cell: ({ row }) => <div cl

```

Figure 28: Upload Zone

4.1.7 Admin: Vector Database Operations

This section describes the maintenance of the **ChromaDB** vector store.

- **Embedding Synchronization:** When a new document is added, the system generates embeddings using the OpenAIEmbeddings model and performs an "Upsert" operation into the vector store.
- **Collection Management:** The dashboard provides tools to "Re-index" or "Clear" specific collections. If a law is repealed, the administrator can delete the corresponding vectors using their source metadata ID, ensuring the AI no longer references outdated legal text.

```

export const columns: ColumnDef<Embedding>[] = [
  {
    header: "vector id",
    cell: ({ row }) => <div className="font-mono text-xs">{row.getValue("id")}</div>,
  },
  {
    accessorKey: "source",
    header: "Source Chunk",
    cell: ({ row }) => <div className="font-medium text-sm truncate max-w-[200px]">{row.getValue("source")}</div>,
  },
  {
    accessorKey: "tokens",
    header: "Tokens",
    cell: ({ row }) => <div className="text-sm">{row.getValue("tokens")}</div>,
  },
  {
    accessorKey: "created",
    header: "Indexed",
    cell: ({ row }) => <div className="text-muted-foreground text-sm">{row.getValue("created")}</div>,
  },
  {
    id: "actions",
    cell: () => {
      return (
        <Button variant="ghost" size="icon" className="h-8 w-8">
          <ArrowUpRight className="h-4 w-4" />
          <span className="sr-only">View</span>
        </Button>
      )
    },
  },
]

export function EmbeddingsTable() {
  const table = useReactTable({
    data,
    columns,
    getCoreRowModel: getCoreRowModel()
  })
}

```

Figure 29: Vector store

4.1.8 System Auditing and Audit Logs

Given the sensitivity of legal advice, every action within the system is strictly logged for accountability.

- **Activity Logging:** The system captures three levels of logs in a dedicated `audit_logs` collection:
 - **Auth Logs:** Login attempts and session revocations (via Better Auth).
 - **RAG Logs:** Every query-response pair, including the specific "Chunks" retrieved from the vector DB. This helps developers identify if the AI is "hallucinating" or if the retrieval was poor.
 - **Admin Logs:** Records of who added, modified, or deleted a legal document.

- **Immutability:** Audit logs are "Append-Only." Once a log is written to MongoDB, it cannot be modified or deleted through the standard UI, ensuring a reliable trail for legal compliance.

```

export function AuditLogTable() {
  const [sorting, setSorting] = React.useState<SortingState>({})
  const [columnFilters, setColumnFilters] = React.useState<ColumnFiltersState>({})
  const [columnVisibility, setColumnVisibility] = React.useState<VisibilityState>({})
  const [rowSelection, setRowSelection] = React.useState({})

  const table = useReactTable({
    data,
    columns,
    onSortingChange: setSorting,
    onColumnFiltersChange: setColumnFilters,
    getCoreRowModel: getCoreRowModel(),
    getPaginationRowModel: getPaginationRowModel(),
    getSortedRowModel: getSortedRowModel(),
    getFilteredRowModel: getFilteredRowModel(),
    onColumnVisibilityChange: setColumnVisibility,
    onRowSelectionChange: setRowSelection,
    state: {
      sorting,
      columnFilters,
      columnVisibility,
      rowSelection,
    },
  })

  return (
    <React.Fragment>
      <div className="flex items-center py-4 gap-2">
        <Input
          placeholder="Filter actions..."
          value={((table.getColumn("action")?.getFilterValue() as string) ?? "")}
          onChange={(event) =>
            table.getColumn("action")?.setFilterValue(event.target.value)
          }
          className="max-w-sm"
        />
        <DropdownMenu>
          <DropdownMenuTrigger asChild>
            <Button variant="outline" className="ml-auto">
              Columns <ChevronDown className="ml-2 h-4 w-4" />
            </Button>
          </DropdownMenuTrigger>
          <DropdownMenuContent align="end">
            <table>
              <tr>
                <th>getAllColumns()</th>
                <th>filter((column) => column.getCanHide())</th>
                <th>map((column) => {</th>
                <th>return {</th>
                <th><DropdownMenuCheckboxItem</th>
                <th>key={column.id}</th>
                <th>className="capitalize"</th>
                <th>checked={column.getIsVisible()}</th>
                <th>onCheckedChange={(value) =></th>
            </table>
          </DropdownMenuContent>
        </DropdownMenu>
      </div>
    </React.Fragment>
  )
}

```

View 1 edited file

```

export function AuditLogTable() {
  // TABLE COLUMN DEFINITION
  .map((column) => {
    return (
      <DropdownMenuCheckboxItem
        key={column.id}
        className="capitalize"
        checked={column.getIsVisible()}
        onChange={(value) => {
          column.toggleVisibility(!value)
        }}
      />
      {column.id}
    )
  })
  </DropdownMenuContent>
</DropdownMenu>
</div>
<div className="rounded-md border overflow-x-auto">
  <Table>
    <TableHeader>
      {table.getHeaderGroups().map((headerGroup) => (
        <TableRow key={headerGroup.id}>
          {headerGroup.headers.map((header) => {
            return (
              <TableHead key={header.id}>
                {header.isPlaceholder
                  ? null
                  : flexRender(
                      header.column.columnDef.header,
                      header.getContext()
                    )}
              </TableHead>
            )
          })}
        </TableRow>
      ))}
    </TableHeader>
    <TableBody>
      {table.getRowModel().rows?.length ? (
        table.getRowModel().rows.map((row) => (
          <TableRow
            key={row.id}
            data-state={row.getIsSelected() && "selected"}
          >
            {row.getVisibleCells().map((cell) => (
              <TableCell key={cell.id}>
                {flexRender(
                  cell.column.columnDef.cell,
                  cell.getContext()
                )}
              </TableCell>
            ))}
          </TableRow>
        ))
      ) : null}
    </TableBody>
  </Table>

```

Figure 30: Audit log

Chapter Five: Testing and Evaluation

5.1 Introduction

This chapter presents the testing and evaluation activities carried out for the **Amharic Legal RAG Chatbot**. Systematic validation is essential to ensure that the system is reliable, correct, and prepared to handle sensitive legal inquiries with accuracy. The phase focused on verifying functional correctness, evaluating system responsiveness under load, and assessing the effectiveness of the Retrieval-Augmented Generation (RAG) pipeline.

Because the system integrates several specialized components—Node.js for orchestration, Flask for AI logic, and a dual-database setup (MongoDB and ChromaDB)—the testing approach was multifaceted. The evaluation combined traditional software testing practices with AI-focused validation using metrics like Faithfulness and Relevancy.

The assessment is organized into the following parts:

- **Functional Testing:** Verification of core features such as Amharic text processing, legal query retrieval, chat history persistence, and administrator-led PDF ingestion.
- **Performance Testing:** Measurement of API latency, vector search speed, and the resource consumption of the Flask/AI container.
- **Security Testing:** Validation of Role-Based Access Control (RBAC), JWT authentication flows, and sanitization of user inputs to prevent prompt injection.
- **AI Evaluation:** Qualitative and quantitative checks for retrieval accuracy (Context Precision) and the grounding of responses (Faithfulness).

5.2 Functional Testing

Functional testing confirmed that all system features behave as expected and meet their legal research requirements. This phase emphasized validating core workflows across the web interface and the backend logic.

5.2.1 Authentication & User Management

A key functional requirement is secure user access. We validated the login flow by sending a POST request to the `/api/auth/login` endpoint. The Node.js server successfully issued a JWT token upon valid credential submission, granting access to protected chat features.

5.2.2 Legal Query & RAG Pipeline

To validate the retrieval process, a legal query in Amharic was submitted. The system successfully executed the following steps:

1. Flask converted the Amharic text into an embedding.
2. ChromaDB returned the top- k relevant legal document snippets.
3. The Gemini Pro API generated a response cited with the correct article numbers.

5.2.3 Administrative Ingestion

Admin controls were tested by uploading a new legal PDF. The system extracted the text, chunked it, and indexed the vectors into ChromaDB, making the new law searchable within seconds.

Table 21: Test Cases

Test Case	Description	Input	Expected Output	Status
TC-01	User Login	Valid Credentials	200 OK + JWT Token	Pass
TC-02	Amharic Query	"የኢትዮጵያ የንግድ ህግ..."	Response	Pass
TC-03	PDF Upload	Labor Law PDF	201 Created + Indexed	Pass
TC-04	History Persistence	Close/Open Session	Past chats displayed	Pass

5.3 Performance Testing

Performance testing evaluated the system's stability and speed. For a RAG application, the bottleneck is often the vector search and LLM inference time.

5.3.1 API and Retrieval Latency

Observations showed that the semantic search in **ChromaDB** remained extremely fast (under 150ms), while the end-to-end response time (including Gemini Pro generation) averaged between **1.5 to 3 seconds**, which is within acceptable limits for a conversational AI.

5.3.2 Container Resource Usage

Monitoring the Docker containers showed stable memory consumption. The Flask container (AI Engine) utilized more CPU during embedding generation, while the Node.js container maintained a low footprint.

5.4 Security Testing

Security testing verified that the system protects legal data and enforces access boundaries.

- **Role-Based Access Control (RBAC):** We validated that unauthenticated users are blocked from the chat interface and that standard users cannot access the /admin ingestion dashboard.
- **JWT Integrity:** Tampered JWT tokens were rejected by the Node.js middleware, preventing session hijacking.
- **Input Sanitization:** We tested various "jailbreak" prompts in Amharic to ensure the system adhered to its system instructions and did not leak internal document metadata.

5.5 AI Evaluation (RAG Accuracy)

Using the **Ragas framework** metrics, we evaluated the quality of the AI's legal advice:

- **Faithfulness:** Measured how well the answer was grounded in the retrieved legal text. The system scored **0.89/1.0**, indicating low hallucination.
- **Answer Relevancy:** Measured how directly the AI answered the user's specific legal question.
- **Context Precision:** Confirmed that the most relevant legal articles were ranked highest in the retrieval results.

5.6 Conclusion

The testing conducted across functional, performance, and security domains confirms the readiness of the **Amharic Legal RAG Chatbot**. The results indicate that the platform provides reliable, cited, and legally grounded assistance. Future iterations will focus on expanding the legal corpus and further reducing LLM inference latency.

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Appendix

Appendix A: Evaluation Data & RAG Metrics

This section details the performance of the AI across various legal domains. We evaluated the system using a test set of **50 Amharic legal queries**.

- **Constitutional Domain:** The system achieved a **Faithfulness score of 0.95** and perfect **Context Precision**, with an average response time of **1.8 seconds**.
- **Labor Law Domain:** In complex scenarios involving the 1156/2011 Proclamation, the system maintained a **0.88 Faithfulness score** with a response time of **2.4 seconds**.
- **Criminal Code Domain:** Retrieval for specific penal articles showed a **0.82 Faithfulness score** and **0.85 Precision**, typically responding within **3.1 seconds**.
- **Commercial Domain:** Queries regarding business registration and trade laws resulted in a **0.91 Faithfulness score** and **0.95 Precision**, averaging **2.2 seconds**.

The overall system average across all tested legal domains reflects a **0.89 Faithfulness score**, a **0.92 Context Precision**, and a mean response time of **2.37 seconds**.

Appendix B: API Endpoints & Data Schema

Detailed documentation for the backend communication between the **Node.js Orchestrator** and the **Flask AI Engine**.

B.1 Core Endpoints

- **POST /api/chat/query:** This is the main entry point for user queries. It requires a valid JWT in the authorization header and handles the primary RAG pipeline request.
- **GET /api/history/{threadId}:** This endpoint fetches past conversation logs and message metadata directly from the MongoDB store for a specific user session.
- **POST /api/admin/ingest:** An administrative-only endpoint used for uploading, chunking, and vectorizing new PDF legal documents into ChromaDB.

B.2 Database Object Sample (MongoDB)

The following JSON structure represents a typical chat thread stored in the metadata layer:

- **Thread ID:** ch_7721983
- **User ID:** usr_992
- **Message 1 (User):** "የሰራተኛ ስንብት ከፍታ ስንት ነው?" (Sent at 2026-01-25T10:00:00Z)
- **Message 2 (Assistant):** "በኢትዮጵያ የሰራተኛ አዋጅ ቁጥር 1156/2011 መሰረት..."

- **Citations:** Includes "Labor Proclamation 1156/2019, Art 39" to ensure transparency.

Appendix C: User Manual & Admin Guide

This appendix serves as a quick-start guide for the primary roles within the system.

C.1 For Legal Researchers (Users)

1. **Search:** Users simply enter their legal question in plain Amharic into the chat interface.
2. **Verify:** Every AI-generated response includes a "Sources" section. Users should click this to see the original law snippet and verify the article numbers.
3. **Manage History:** The sidebar allows users to revisit past research, rename chat threads for better organization, or delete sessions they no longer need.

C.2 For System Administrators

1. **Accessing the Dashboard:** Admins navigate to the `/admin` route and authenticate with their specific administrative credentials.
2. **Knowledge Base Updates:** New official gazettes (Negarit Gazeta) are added by dragging and dropping PDFs into the upload zone.
3. **Monitoring Indexing:** The system provides a real-time status bar. Once "Indexing Complete" appears, the new legal data is live and available for user queries.